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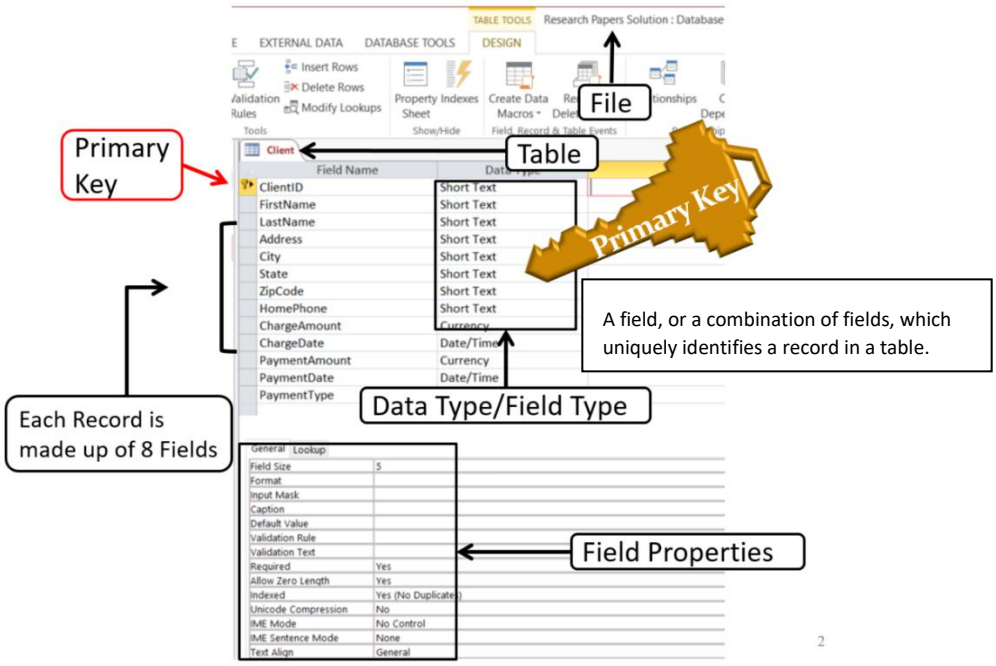
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PART 1: DESIGNING A DATABASE

Read Pages 2-7 of this Lab. Starting on page 8, you will complete the steps to create a database, using the files you download from Simnet.

WHAT IS A DATABASE?

A database is used to store information about a person, place, or thing. Almost all information retrieved from a computer is stored in a database. There are several different types of databases or what we call database models. One of the database models used is called the relational database model. This model stores information in files, tables, records, and fields.



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STEPS WHEN CREATING A DATABASE

There are several steps that should be taken when creating a new database. These steps should usually be in the following order, although sometimes that may not be the case.

Step 1 – Design and Document the database

The first four steps when creating a database entail documenting and designing the database. This document will serve as a map for creating the actual database itself.

1. Decide what information you will store in the database and document it on paper*.
2. Create a layout of the database and document it on paper*.
3. Normalize the database and document it by using software for that purpose.
4. Create a Relationship Diagram on paper*.
    a. Identify the Tables, Relationships, Primary Keys and Foreign Keys.

On Paper means to design and document the database structure (using software for that purpose or even MS Word) before creating the database.

Step 2 – Create the database using MS Office Access or any other database software.

The final steps entail the creation and setup of the database using the database software. *(In this lab we will be using MS Access to create the database.)*

5. Create the database structure in MS Access.
6. Create the table relationships
7. Enforce Referential Integrity in MS Access.
8. Populate the MS Access database.
    a. This entails typing the information into the newly created database.

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STEP 1 – DECIDE WHAT INFORMATION YOU WILL STORE IN THE DATABASE.

One of the first steps in creating a database is to decide what information you will store in the database. For example, let's say you have started a small business using MS Word to type research papers for the students at your school. You want to create a database to store all your charges and payments so you can collect your fees in a timely manner. What information do you need to store and how will you make the information for each person unique? For the purpose of this example, we will only store the minimum amount of information needed to keep track of your charges and payments received. ***Figure 1.1*** shows the basic information needed to create this database.

First Name	Charge Amount
Last Name	Charge Date
Address	Payment Amount
City	Payment Type
State	Payment Date
Zip Code	
Home Phone	

Figure 1.1

This database is a very simplistic one. Most databases you create will be much more complex. Therefore, when you are collecting the information needed for your database, you should always consult the people who actually do the work. You must know the business in order to create the database, so involve the people who do the business!

STEP 2 – CREATE THE LAYOUT OF YOUR DATABASE USING SOFTWARE FOR THAT PURPOSE

Once you have decided the information to be included in your database, you must now create the layout of the database. The layout consists of the tables, records, fields, field types, and field properties that will be included in your database. Basically, this process of designing a solid and accurate database is called **Normalization.** In order to normalize the database, you must group related fields into tables, while eliminating any duplicate data in the database except the ***primary keys*** and ***foreign keys*** which will relate the tables together. You also must create a primary key for most of the tables in your database. What is a primary key? A ***primary key*** is the one field that will make each record unique in your table. *(Note: You can create a primary key from more than one field, but that is beyond the scope of this book.)* Have you ever ordered a pizza? What is the first question usually asked when you call…Could I have your home phone number please? It's the home phone number that makes you unique as a Client. Your social security number makes you unique in the United States, and your Student ID makes you unique at most Universities. We will create a unique field for each of our clients. ***Figure 1.2*** displays the table I have created from the information I have decided I need to store in ***Figure 1.1***. I have created one table called Client, which will store all of the personal information needed for each client, and all the charges and payments for each client. Each piece of this information is called a field, and all of the fields combined for one person constitute one record. Hence, if I have 3 clients in the table in ***Figure 1.2***, I have 3 records and 39 fields. ***This database has not been normalized.***

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Research Papers Database (Not Normalized)

Table Name:	Client	Field Type	Properties (Field Size)	Notes	
Fields:	ClientID	Short Text	5	Primary Key	
	FirstName	Short Text	25		
	LastName	Short Text	50		
	Address	Short Text	60		
	City	Short Text	25		
	State	Short Text	2		
	ZipCode	Short Text	5		
	HomePhone	Short Text	10		
	ChargeAmount	Currency	None		
	ChargeDate	Date/Time	None		
	PaymentAmount	Currency	None		
	PaymentDate	Date/Time	None		
	PaymentType	Short Text	2		

Figure 1.2

Why not just create one big table like the one above and call it a day? The problem is that every time a client is charged, or a payment is made, all of the information needs to be re-input for each client. This causes quite a bit of information to be duplicated and could also leave room for inaccurate information. What if the first record has our Client listed as Rebecca Simpson, and the 2nd record has her listed as Becky Simpson? Now our information is incorrect. Are Rebecca Simpson and Becky Simpson the same people? What if we have a Rebecca Simpson at PO Box 189 and a Rebecca Simpson at PO Box 198? Easy, you say, just look at the database to see if the information looks the same. That is fine with a small database, but what if you have a database with thousands of records? That would be a waste of time. Another problem this creates is wasted disk space because we are duplicating information over and over.

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STEP 3 -NORMALIZE THE DATABASE

As previously stated, in order to normalize the database, you must group related fields into tables while eliminating any duplicate data in the database except the primary keys and foreign keys which will relate the tables together. You also must create a primary key for most of the tables in your database.
Figure 1.3 shows the database normalized using documentation software. (There are many different forms of Normalization. This book will not detail the different forms.) This handout will not detail the different forms of normalization.)

DOCUMENTING THE RESEARCH PAPERS DATABASE STRUCTURE (NORMALIZED)

Table Name:	**Client**	**Field Type**	**Properties**	
Fields:	ClientID	Short Text	5 Characters long	Primary Key
	FirstName	Short Text	25 Characters long	
	LastName	Short Text	50 Characters long	
	Address	Short Text	60 Characters long	
	City	Short Text	25 Characters long	
	State	Short Text	2 Characters long	Default Value
	ZipCode	Short Text	5 Characters long	
	HomePhone	Short Text	10 Characters long	Input Mask
Table Name:	**Charges**	**Field Type**	**Properties**	
Fields:	ClientID	Short Text	5 Characters long	
	Amount	Currency	None	
	ChargeDate	Date/Time	Input Mask	
Table Name:	**Payments**	**Field Type**	**Properties**	
Fields:	ClientID	Short Text	5 Characters long	
	Amount	Currency	None	
	PaymentType	Short Text	2 Characters long	
	PaymentDate	Date/Time	Input Mask	
Table Name:	**PaymentMethod**	**Field Type**	**Properties**	
Fields:	MethodID	Short Text	2 Characters long	
	MethodType	Short Text	20 Characters long	

Figure 1.3

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There are four tables documented.

1. **Client Table:** Stores all the personal information about the client. Each client will only need one record in this table; thus duplication of information and inaccuracy has been reduced drastically. Notice the primary key is ClientID. This is the field we will use to make each record unique in the Client table.

2. **Charges Table:** This table stores all information on client charges. Because we may have many charges for one client, we will not have a primary key on this table. However, we will need a field that tells us what charge is related to what client. For this we will use the ClientID field. Therefore, for every Client in the Client table, we can have zero to many charges for that Client in the Charges table.

3. **Payments Table:** This table stores all information on client payments. Because we may have many payments for one client, we will not have a primary key on this table. However, we will need a field that tells us what payment is related to what client. For this we will use the ClientID field. Therefore, for every Client in the Client table, we can have zero to many payments for that Client in the Payments table.

4. **PaymentMethod Table:** This table stores all information on payment methods. The client can pay by Visa, Mastercard, Purchase Order (PO), Check, or Cash. We will document this on every payment.

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STEP 4 - CREATE A RELATIONSHIP DIAGRAM TO IDENTIFY THE TABLES, RELATIONSHIPS, PRIMARY KEYS, AND THE FOREIGN KEYS

The tables have been created, but how can the database know what record in the Client table relates to records in the Charges and Payments table. In other words, how can I make sure that I can pull the correct charges and payments information for each client? This is done by creating relationships between the tables; hence we will create a relational database. We will depict this on paper (or documentation software) by creating a **Relationship Diagram** as shown in ***Figure 1.4***.

Depiction of relationship diagram using documentation software.

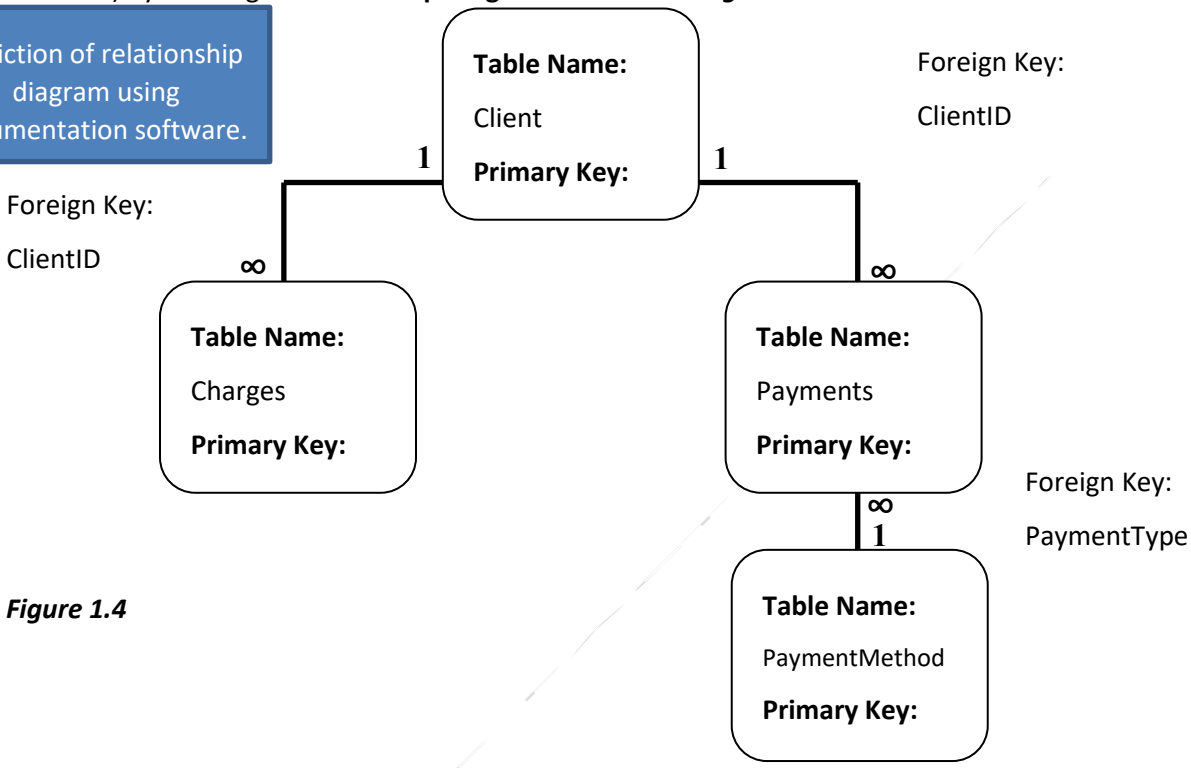


Figure 1.4

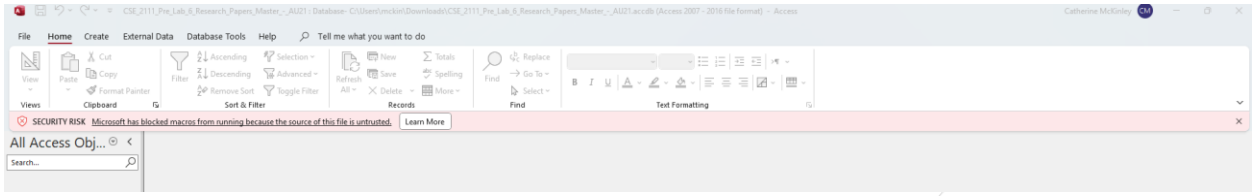
It is obvious the field which makes each record in the Client table unique is the ClientID field, but what about the Charges and Payments tables? These two tables can have many charges and payments from one client; therefore, they will not have a primary key. But how do we relate the records in each table. We use the ClientID field on the Charges table to relate each charge to one client on the client table, and we use the ClientID field on the Payments table to relate each payment to one client on the Client table. Therefore, there is a one-to-many relationship between the Client table and the Charges table, and there is a one-to-many relationship between the Client table and the Payments table. The ClientID field on the Charges table is called a Foreign Key, because it will relate all the records on the Client table which use ClientID as its Primary Key. What would be the Foreign Key on the Payments table that will relate the Payments and Client table? It will be the ClientID on the Payments table.

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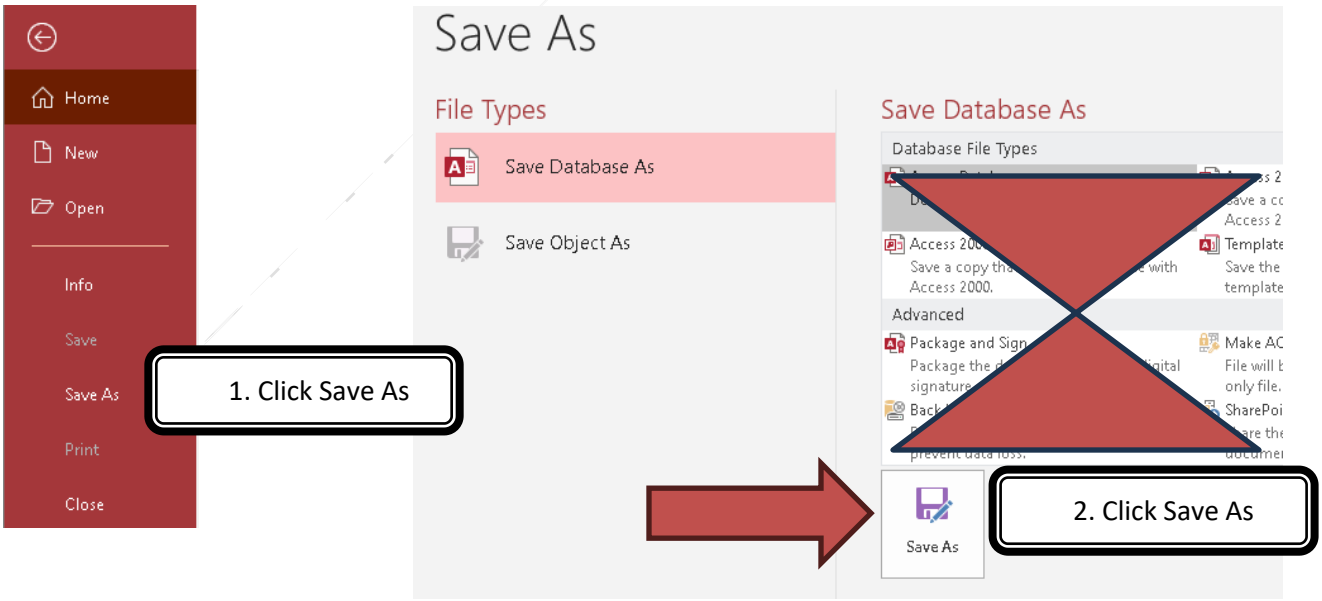
START HERE TO COMPLETE THE LAB

YOU WILL USE THE FILE: CSE 2111 PRE LAB 6 RESEARCH PAPERS MASTER.ACCDB YOU DOWNLOAD FROM SIMNET.

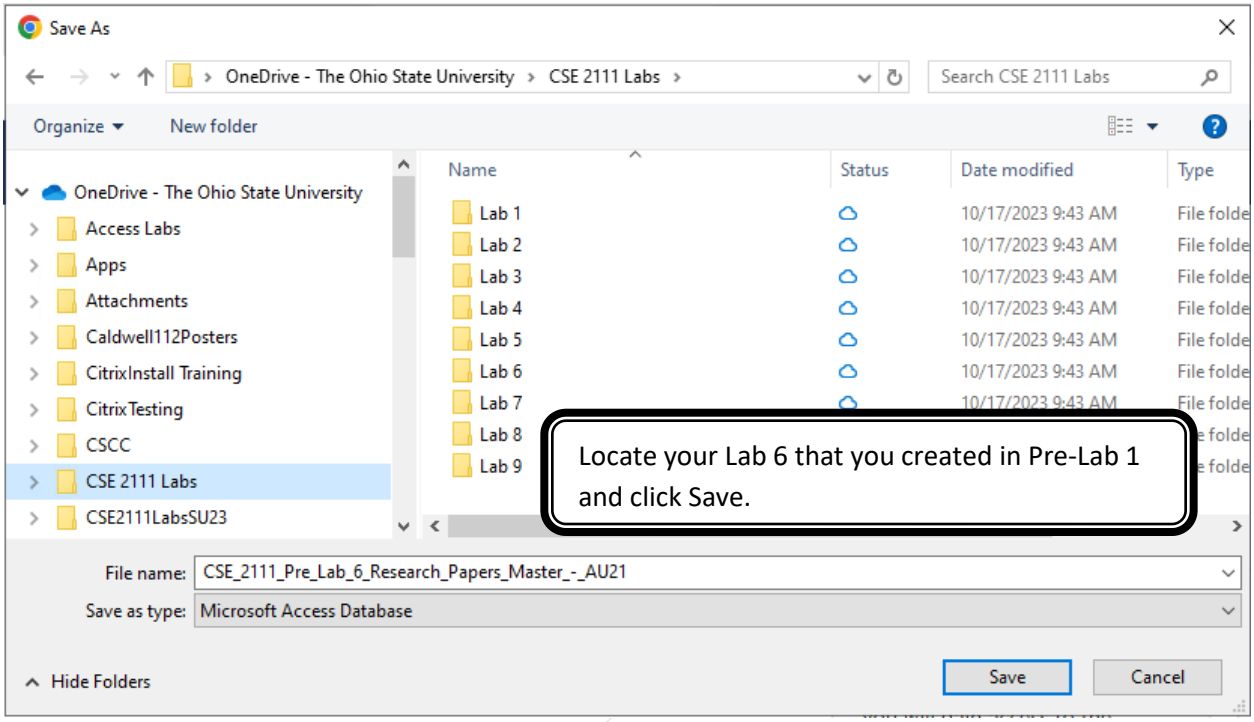
STEP 5 - CREATE THE DATABASE STRUCTURE IN MS ACCESS. OPEN THE CSE 2111 PRE-LAB 6 RESEARCH PAPERS MASTER FILE THAT YOU DOWNLOAD FROM SIMNET. THIS IS A BLANK DATABASE, BUT YOU NEED TO USE THIS FILE.

1. 

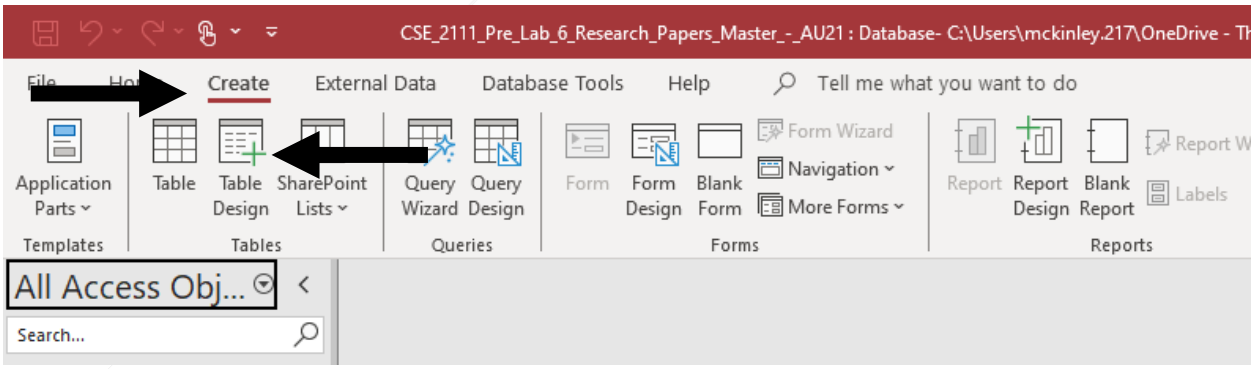
When you open the file, the top part of Access should look like this image. **Ignore the Security Warning. You can click on the 'x' that is on the right side in the pink bar to close the message.**
2. Click the File tab and then click Save As. Then click Save As under Save Database As. (See images below)
3. **YOU NEED TO CLICK THE DISKETTE IMAGE HERE FOR EVERY FILE YOU SAVE - DO NOT CHOOSE ANY OF THE OPTIONS LISTED ABOVE THE DISKETTE IMAGE THAT ARE CROSSED OUT WITH THE BIG RED X.**
4. Save your file in the Lab 6 folder you created in Pre-Lab 1 (if using a PC in a campus lab or in a campus library, save lab files in the OSU OneDrive). See first image on page 9.



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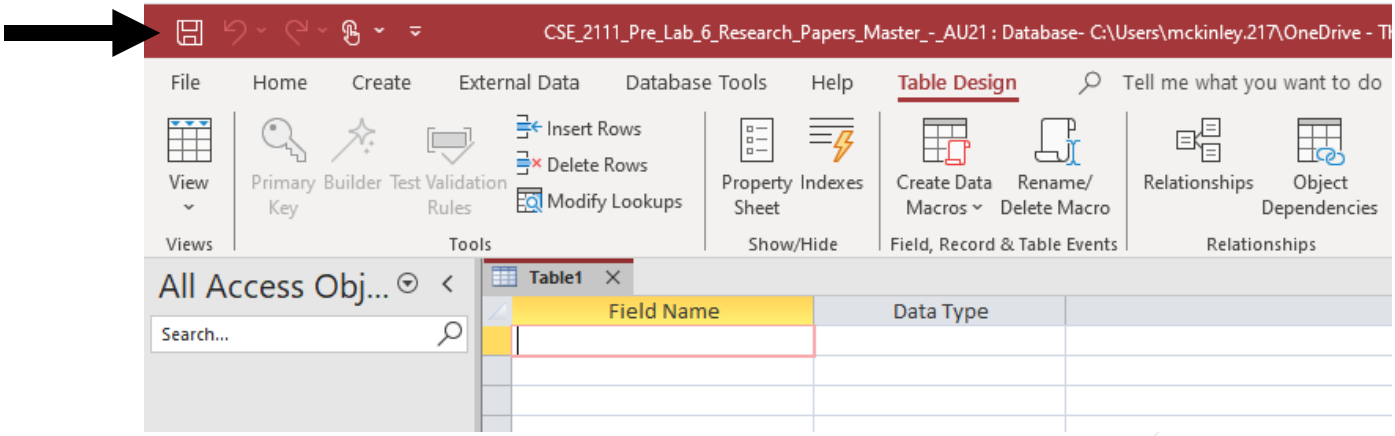
5. Click **Create** and then Click **Table Design in the Tables Group**.



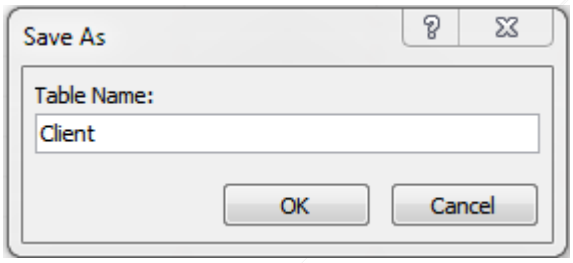
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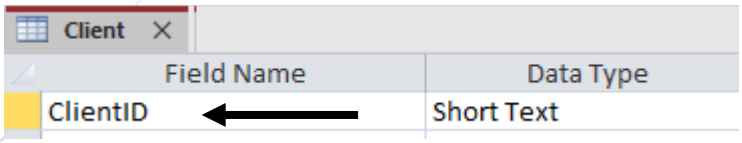
6. Click the Save icon in the upper left corner to display the Save As dialog box.



6. Name the table, **Client. Click OK.**

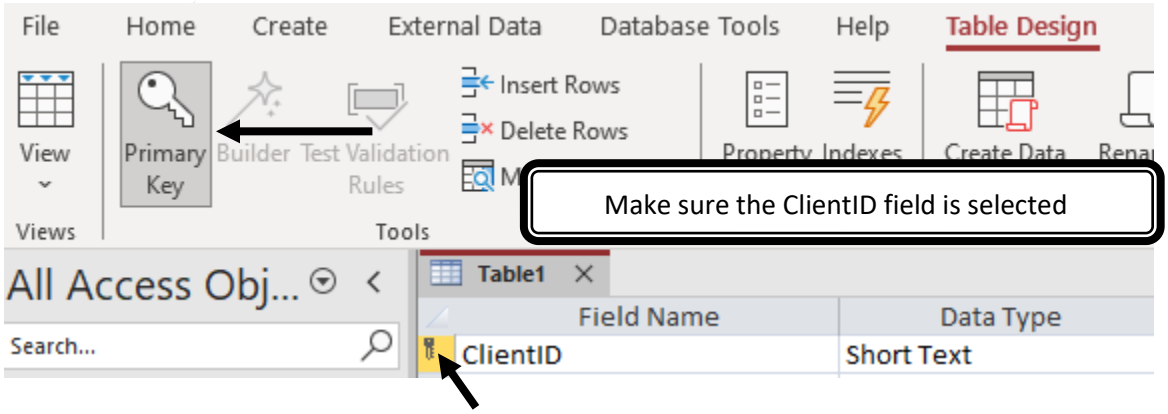


7. Type in the Field Name, ClientID.



(Note that Field Names have no spaces in them.)

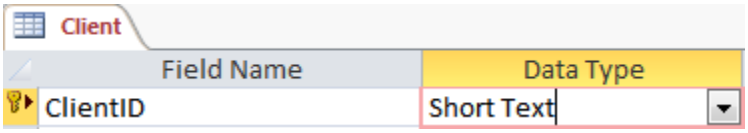
8. Make ClientID the Primary Key by clicking on the Primary Key icon. It will display a key to the left of the ClientID field name.



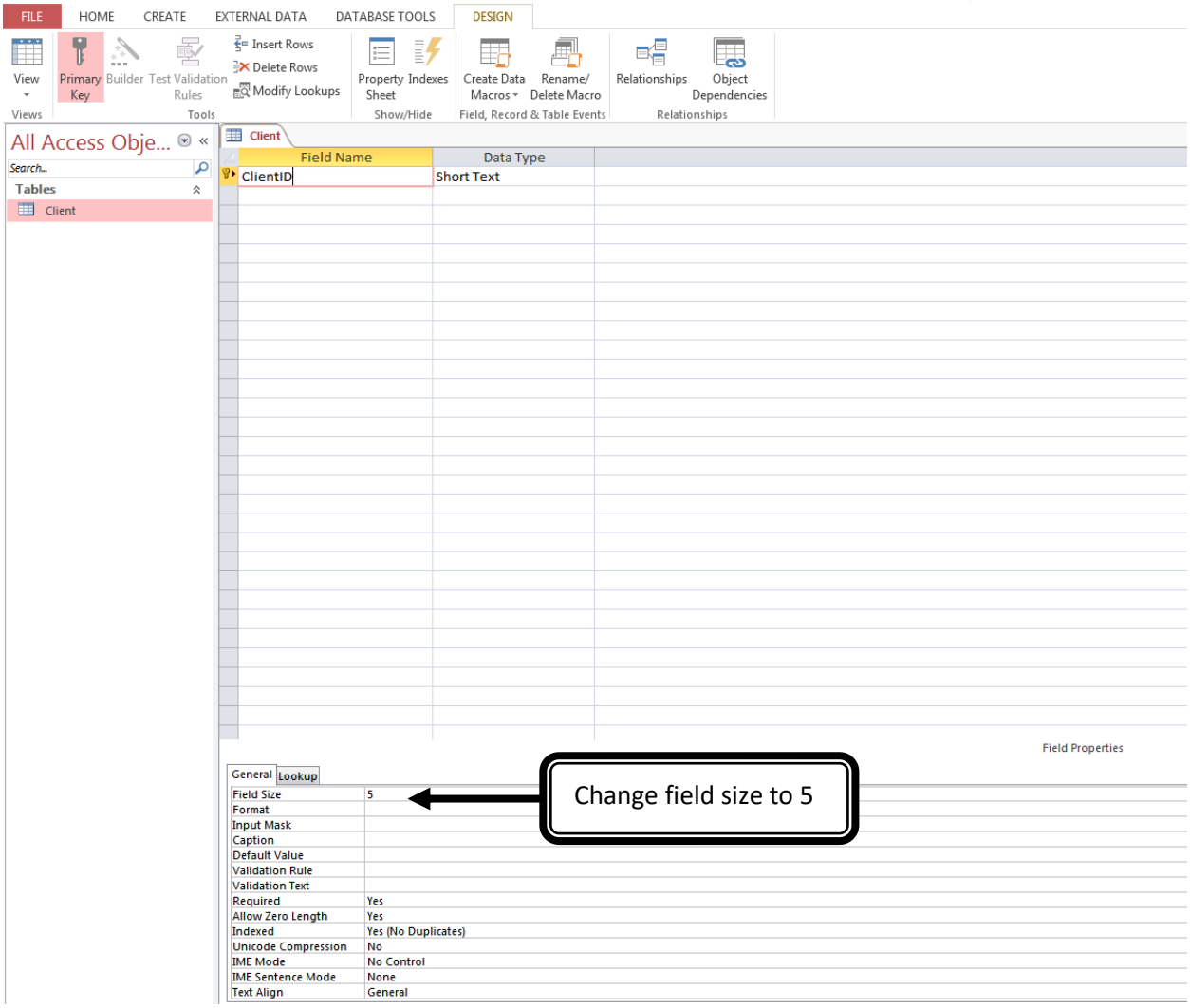
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9. Leave the Data Type as Short Text.

(Note the terms data type and field type can be used interchangeably)



10. Change the Field Size to 5.
    (You must have the ClientID field selected to show the properties for the ClientID field.
    (Note: This will limit the maximum number of characters in the field to 5)
    (Note: Field size is one type of property. Notice all the other properties that can be set)



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NOTE

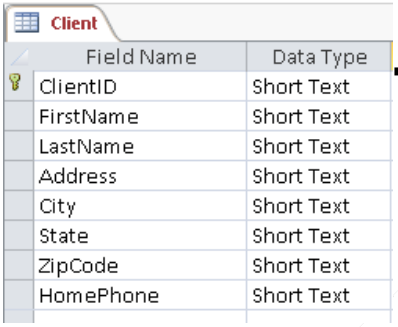
As stated previously, a Primary Key is the one field that will make each record unique in the database table. In this table, ClientID is the field that will make each client's information unique. Notice below that the field ClientID will now have a key next to it. This indicates that you have selected that field as the primary key. You can remove the key by clicking on the field name, and then clicking on the Key icon. You can assign a primary key to a field by clicking on the field and then clicking on the Key icon.

11. Create the remainder of the Client table as shown in ***Figure 1.5*** and close the Client table.

Table Name:	**Client**	**Field Type**	**Properties (Field Size)**		
Fields:	ClientID	Short Text	5	Primary Key	
	FirstName	Short Text	25		
	LastName	Short Text	50		
	Address	Short Text	60		
	City	Short Text	25		
	State	Short Text	2		
	ZipCode	Short Text	5		
	HomePhone	Short Text	10	Input Mask	

Figure 1.5

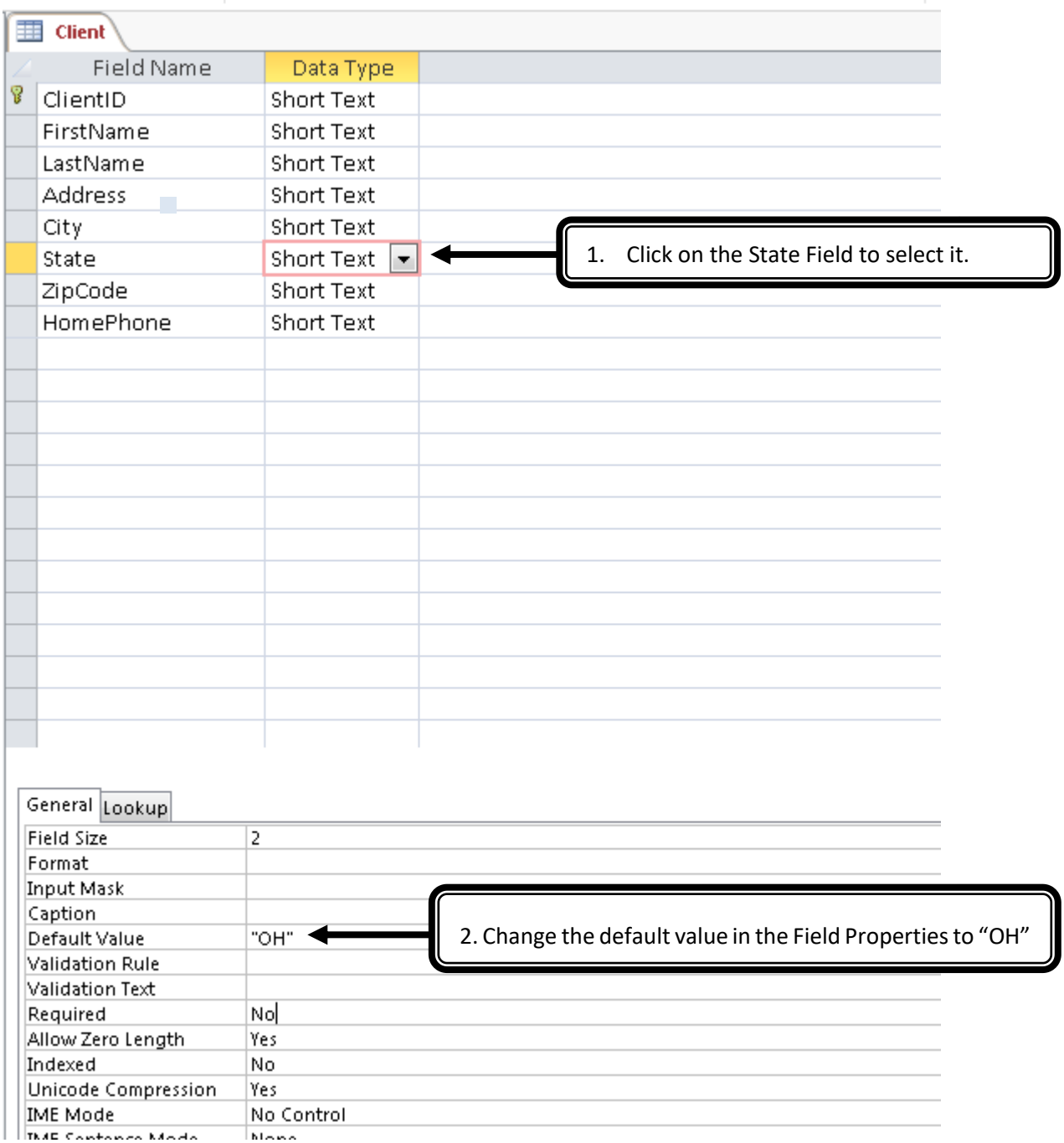
These Field Properties will be added later.



11. Create the fields. Make sure to change the data types and field sizes.

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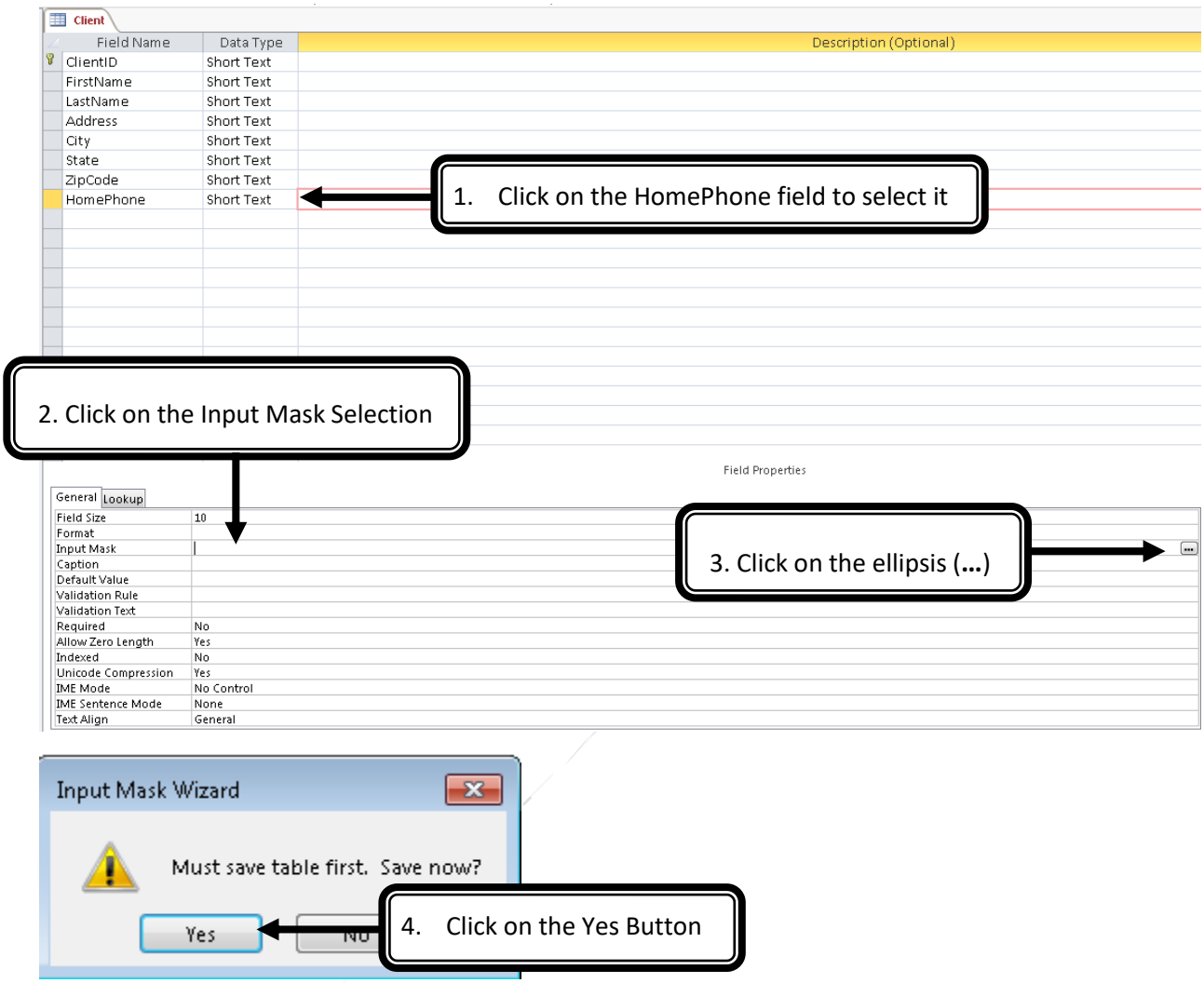
12. Click on the State field in the upper pane to select the field. Change the default value in the field properties to OH.



We have added a default value of ***OH*** to the state field properties. This will cause the state field to initially store the value ***OH*** in every record. However, this value is not set in stone. You can change the value in the field at any time. The default value property is good to use if a certain field will contain the same information in it most of the time. For example, we do most of our business in Ohio, therefore; we have set the default value to ***OH***.

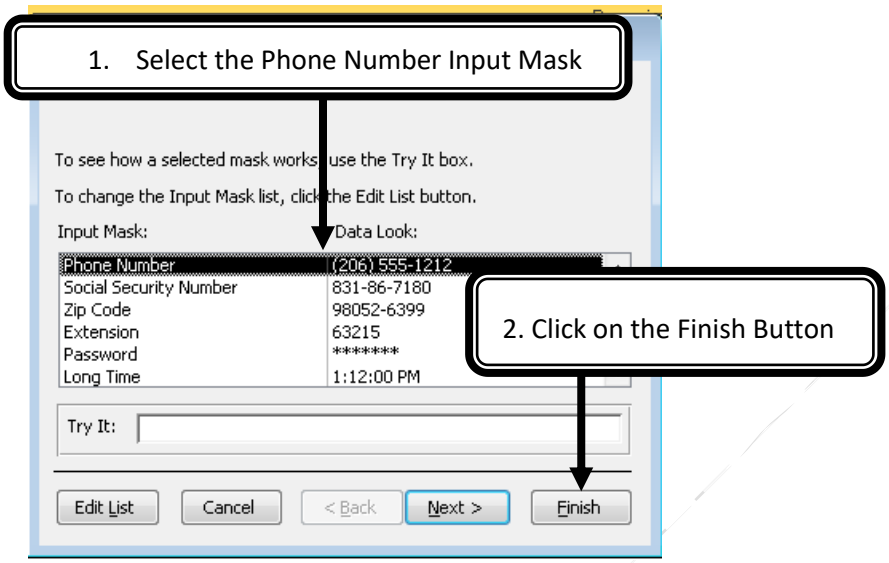
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13. Click on the HomePhone field in the upper pane to select the field. Click on the Input Mask selection in the Field Properties, and then click on the ellipsis (**...**) next to the Input Mask selection.



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14. Click on the Phone Number selection, and then click on Finish.



We have added an input mask of !\(999") "000\-0000;;_ to the HomePhone field property. This will save us the time and hassle of having to add the parentheses and hyphen to the home phone field when we type it in. Basically, the field is preformatted for us. All we have to do is input the actual phone number.

Note: The parentheses and hyphen are not stored in the database. They are included for formatting purposes on the form.

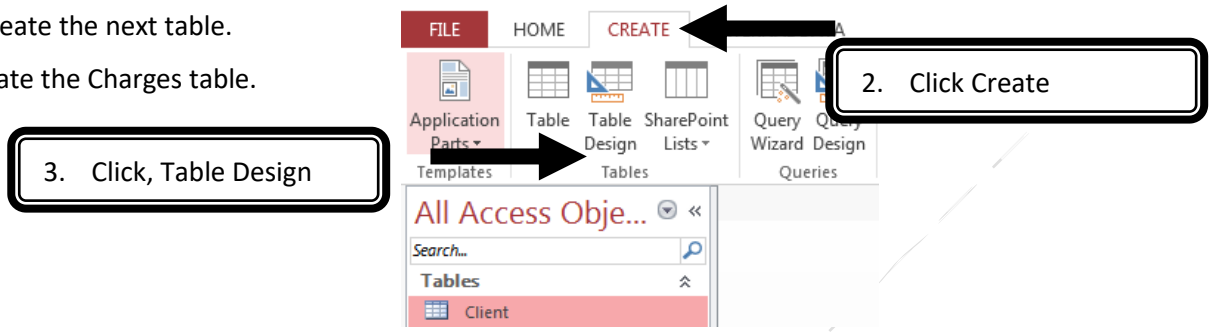
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15. Close and save the table. **Note: DO NOT CLOSE THE ENTIRE DATABASE – ONLY THE TABLE**.



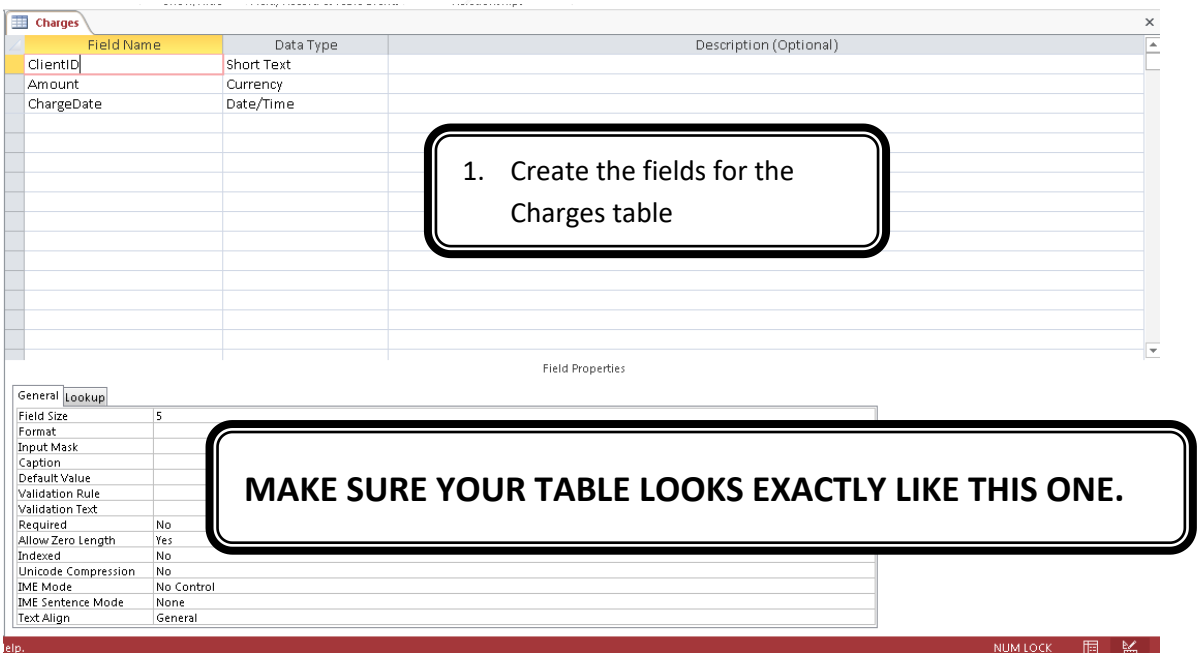
Now, create the next table.

16. Create the Charges table.



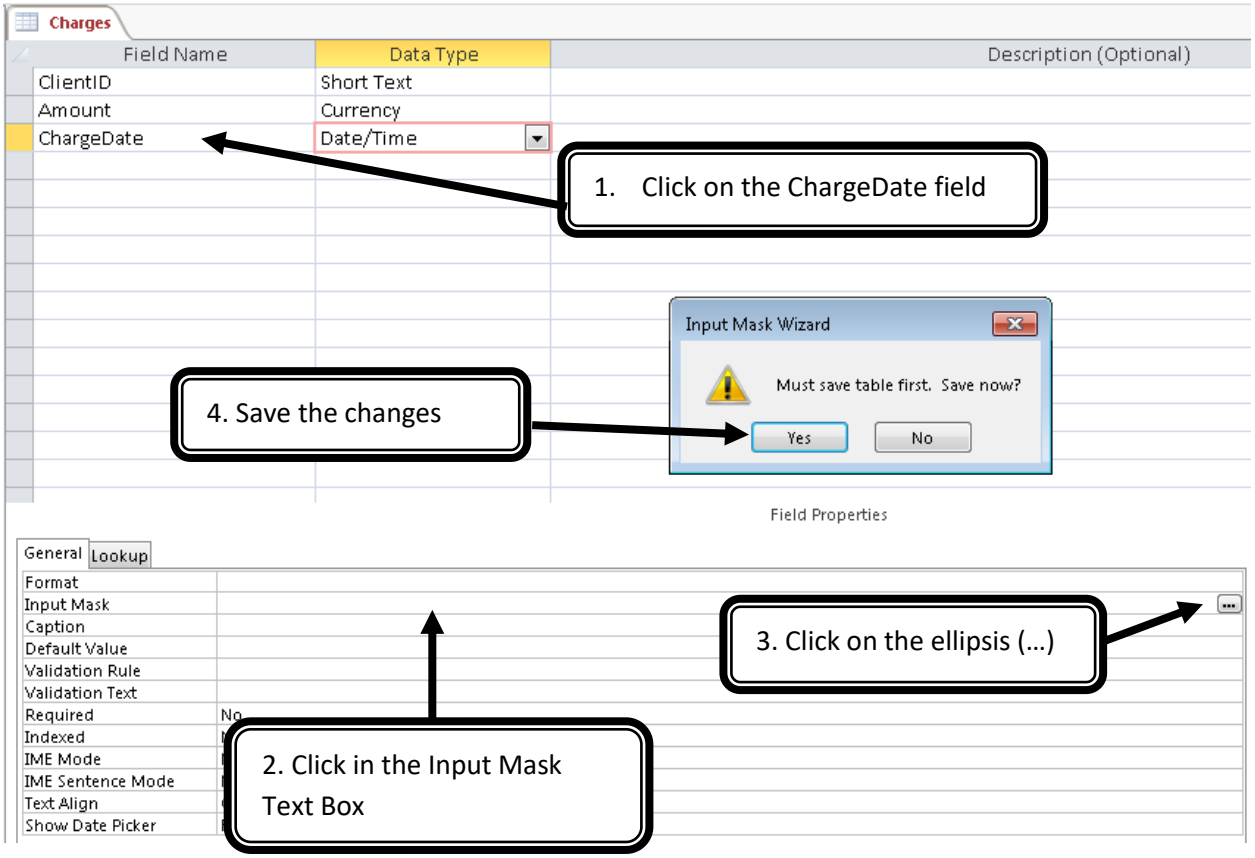
17. Create the structure for the Charges table using the following information.

Table Name:	Charges	Field Type	Field Size (Properties)	
Fields:	ClientID	Short Text	5	
	Amount	Currency	None	
	ChargeDate	Date/Time	Date Input Mask	

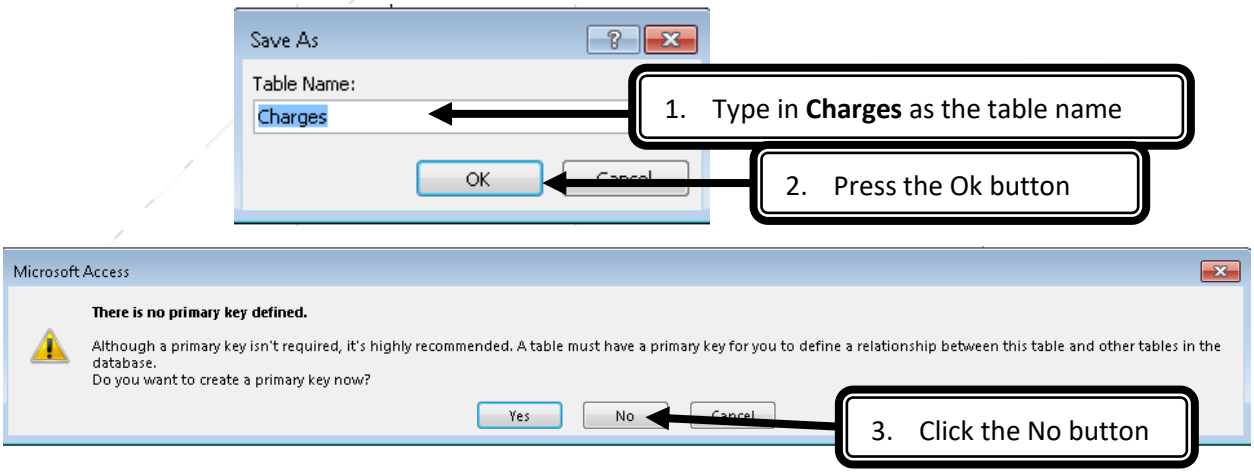


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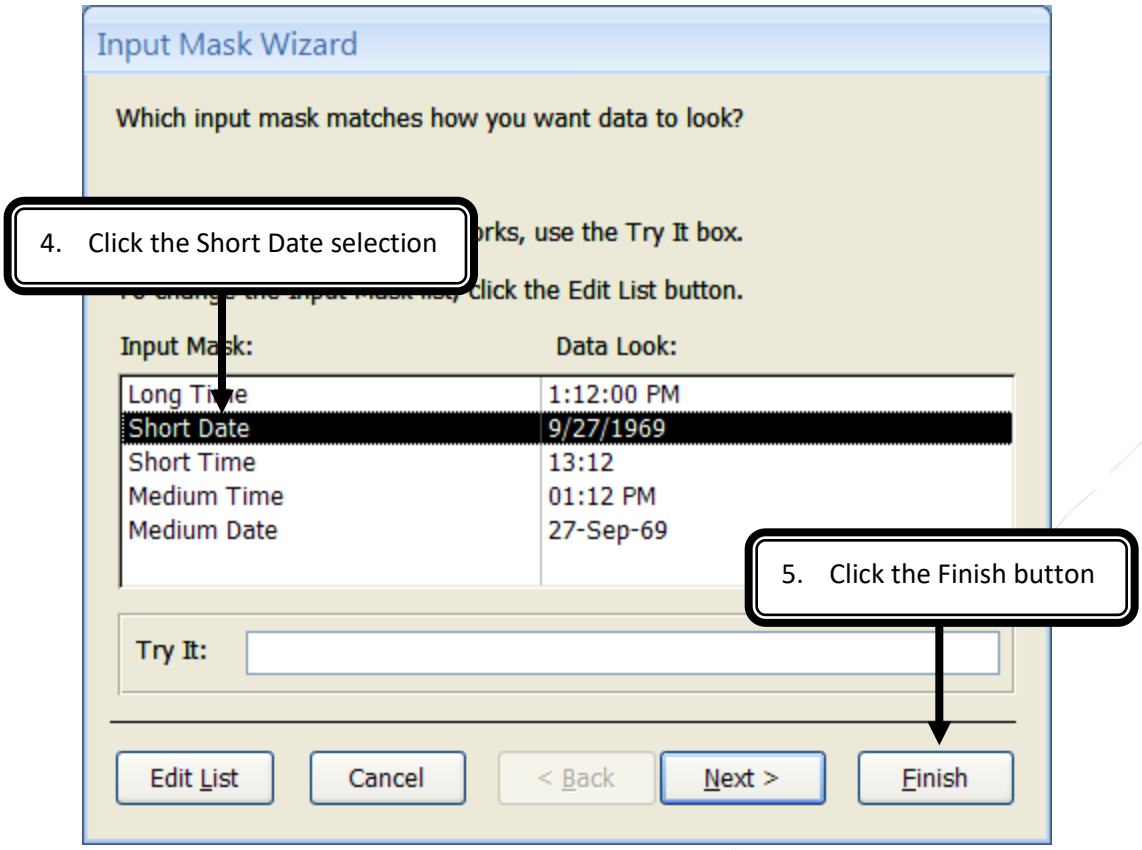
18. Click on the ChargeDate field in the upper pane to select the field. Click on the Input Mask selection in the Field Properties, and then click on the ellipsis (**...**) next to the Input Mask selection. Click the yes button, when asked to Save now.



19. Name the table, Charges and click the OK button. Click the No button when asked, Do you want to create a primary key now? Click on the short date input mask option, and click on finish.

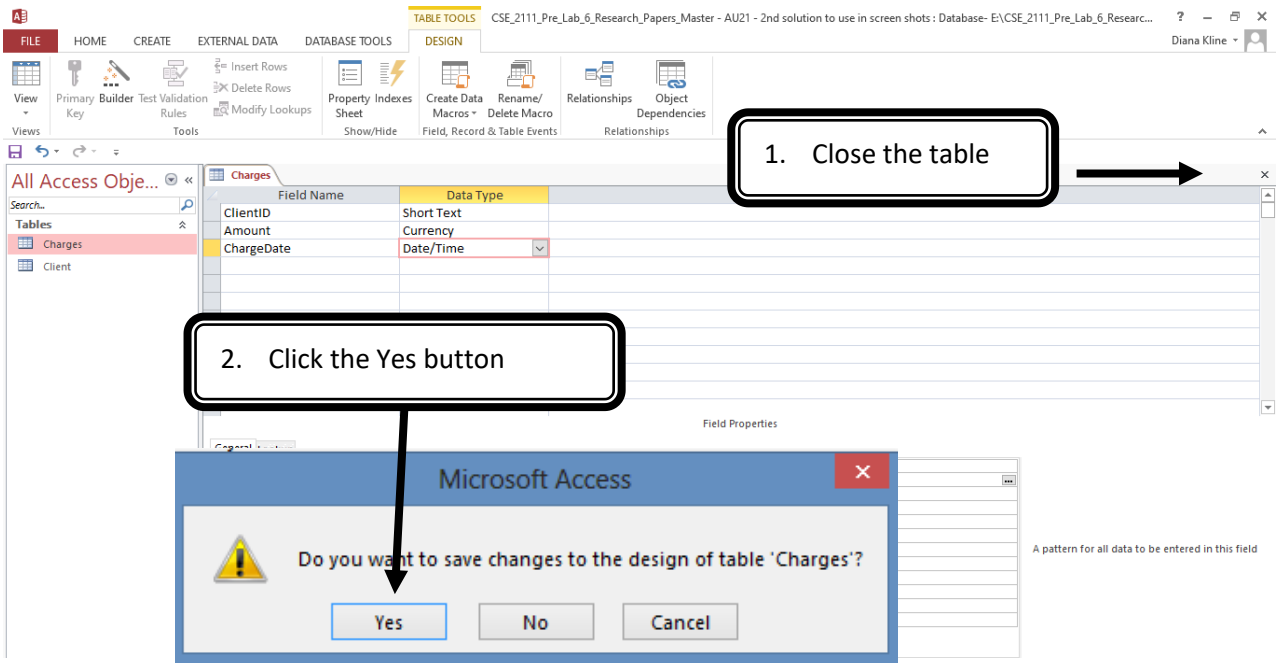


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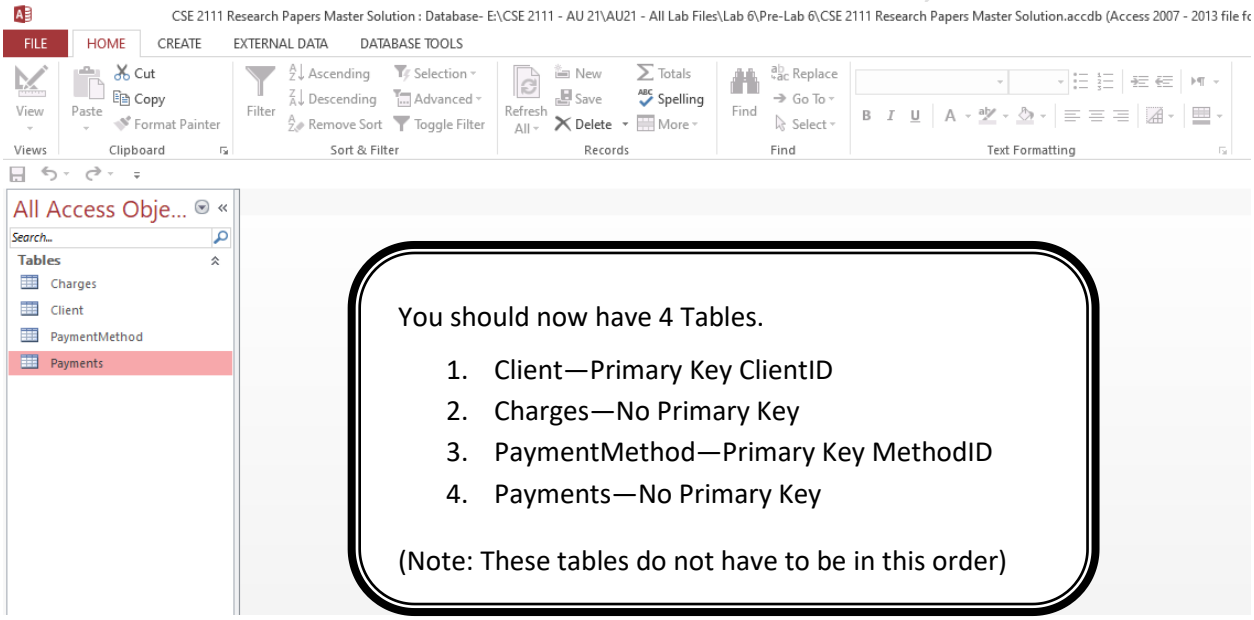
20. Close the table and save the changes.



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21. Follow the same steps to create the Payments table and the PaymentMethod table shown below. Note that the PaymentMethod table was created with a space in the table name – as Payment Method (see screenshot below). Either way is fine. Be sure to create an input mask of short date for the PaymentDate field.

Table Name:	**Payments**	**Field Type**	**Field Size (Properties)**	
Fields:	ClientID	Short Text	5	No Primary Key
	Amount	Currency	None	
	PaymentDate	Date/Time	Input Mask	Short Date
	PaymentType	Short Text	2	
Table Name:	**PaymentMethod**	**Field Type**	**Field Size (Properties)**	
Fields:	MethodID	Short Text	2	Primary Key
	MethodType	Short Text	20	



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STEP 6 – CREATE THE TABLE RELATIONSHIPS

Let's look at the relationship between the Client table and the Charges table. Each record in the Client table has a ClientID field that makes each record unique. Thus, we know if we have a client named, Nancy Davolio and we have assigned her a primary key of J7500, we know J7500 is the field that makes the record for Nancy Davolio unique. How do we know what charges are Nancy's? We must create relationships between the Client table and the Charges table. We will relate the client in the Client table to the client in the Charges table by using the ClientID field on the Client table and the ClientID field on the Charges table. To relate these two tables, both of the tables must have a ClientID field and the ClientID field must be a primary key on one of the tables. The ClientID field in the Client table is a primary key in the Client table. There is also a ClientID field in the Charges table (which is not the primary key in the table) so we can relate these two tables by the ClientID on both tables. Thus, the ClientID on the Client table will be "joined" to the corresponding ClientID on the Charges table. The ClientID on the Charges table is called the foreign key in the relationship.

We have three relationships in this database.

The Client table is related to the Charges table.

- The ClientID in the Client table is the primary key in the relationship, so the Client table is called the parent in this relationship.
- The ClientID in the Charges table is the child/foreign key in the relationship.
- There is a one-to-many relationship between the Client Table and the Charges table
  - For every one record in the Client table there can be zero to many related records in the Charges table

The Client table is related to the Payments table.

- The ClientID in the Client table is the primary key in the relationship, so the Client table is called the parent in the relationship.
- The ClientID in the Payments table is the child/foreign key in the relationship.
- There is a one-to-many relationship between the Client Table and the Charges table
  - For every one record in the Client table there can be zero to many related records on the Charges table

The PaymentMethod table is related to the Payments table.

- The PaymentMethod Table is the parent in this relationship.
- The Payments Table is the child in this relationship.
  - For every one record on the PaymentMethod table there can be zero to many related records in the Payments table

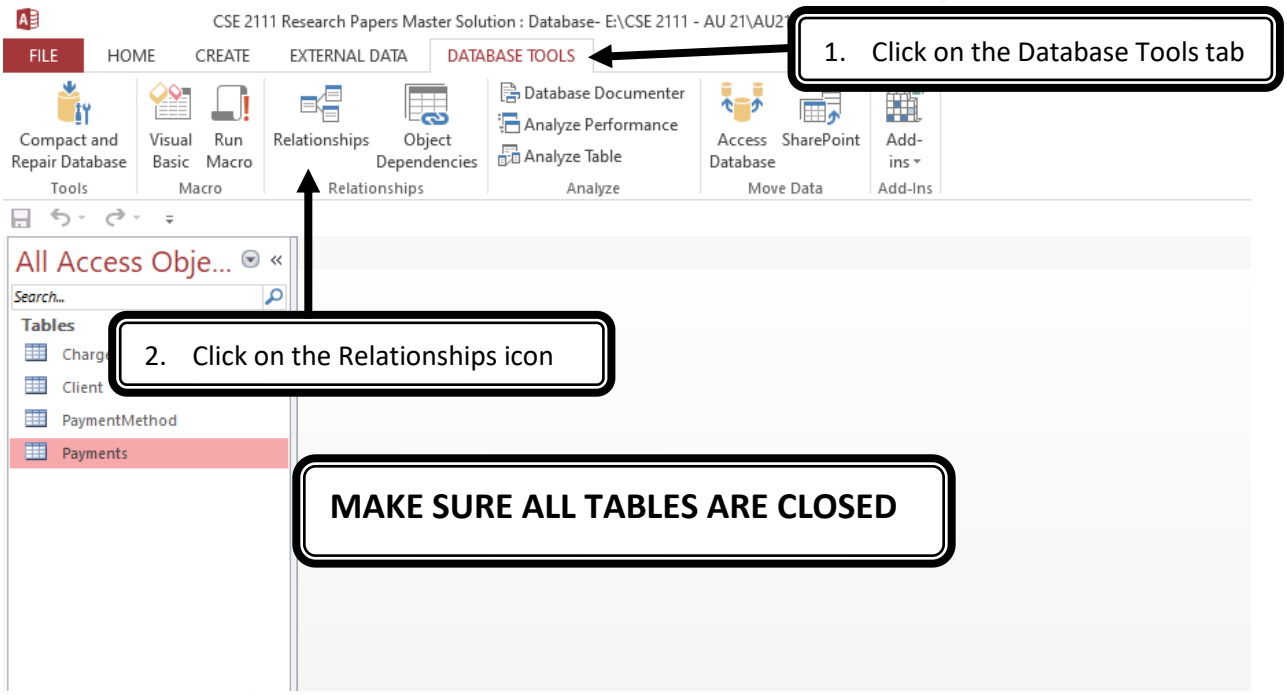
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STEP 7 – ENFORCE REFERENTIAL DATA INTEGRITY

What if we have a record in the Charges table that has no matching record in the Client table? How would we know what Client to charge, or where to send the bill? We won't, and therefore records in the Charges table should always have a matching record in the Client table. Remember the Client table is known as the parent in this relationship, and the Charges table is known as the child. When designing databases, we never want to have orphans in our tables. (A child without a parent.) We can eliminate orphans by enforcing referential data integrity. When referential data integrity is enforced, a child must always have a parent. So, if we try to add record J9900 to the Charges table (which is the child table in the relationship between the two tables) and there is no record J9900 in the Client table (which is the parent table in the relationship) an error message will display letting us know that referential data integrity has been violated, and the record will not be added to the Charges table.

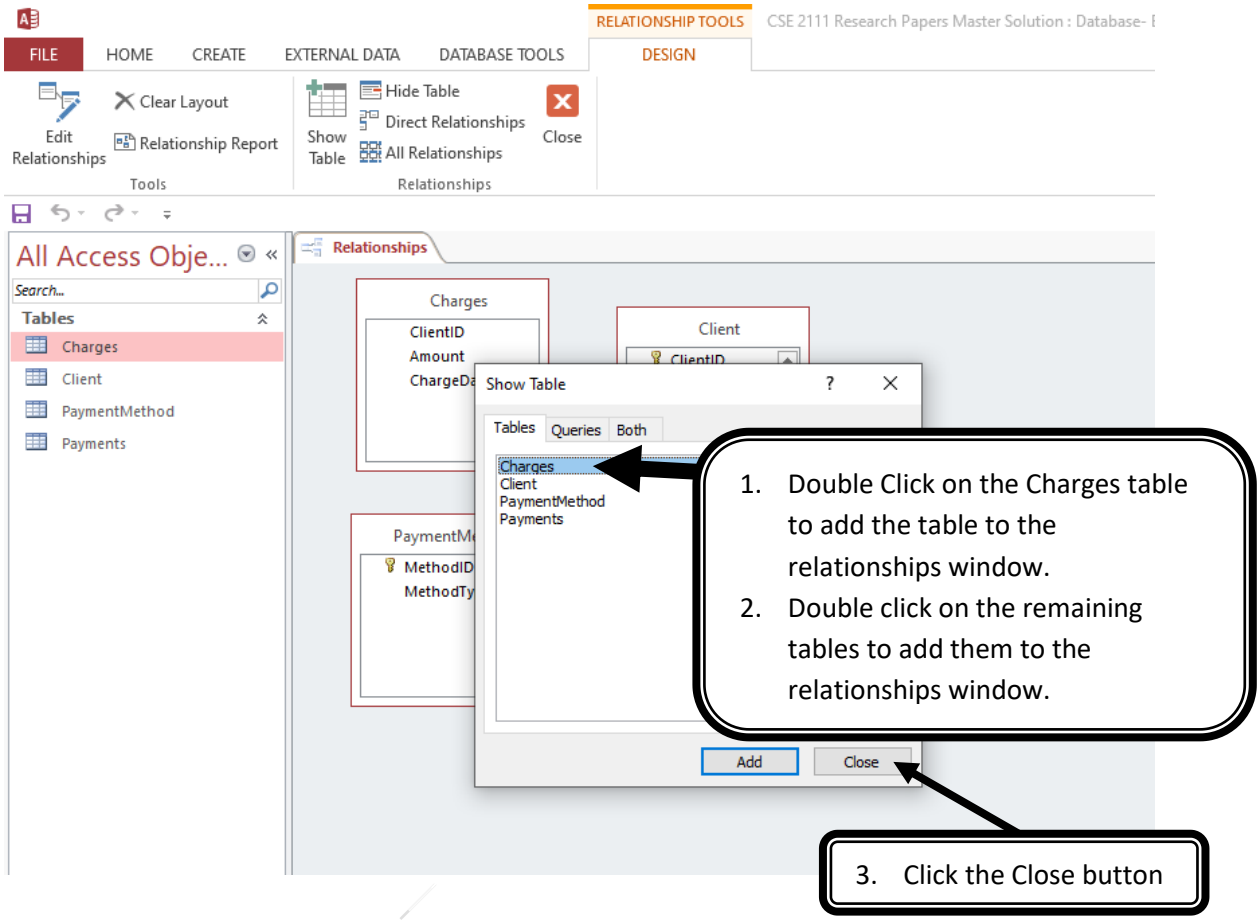
Create the Table Relationships and Enforce Referential Integrity using MS Access.

1. Click on the Database Tools tab, and then click on the Relationships Icon.



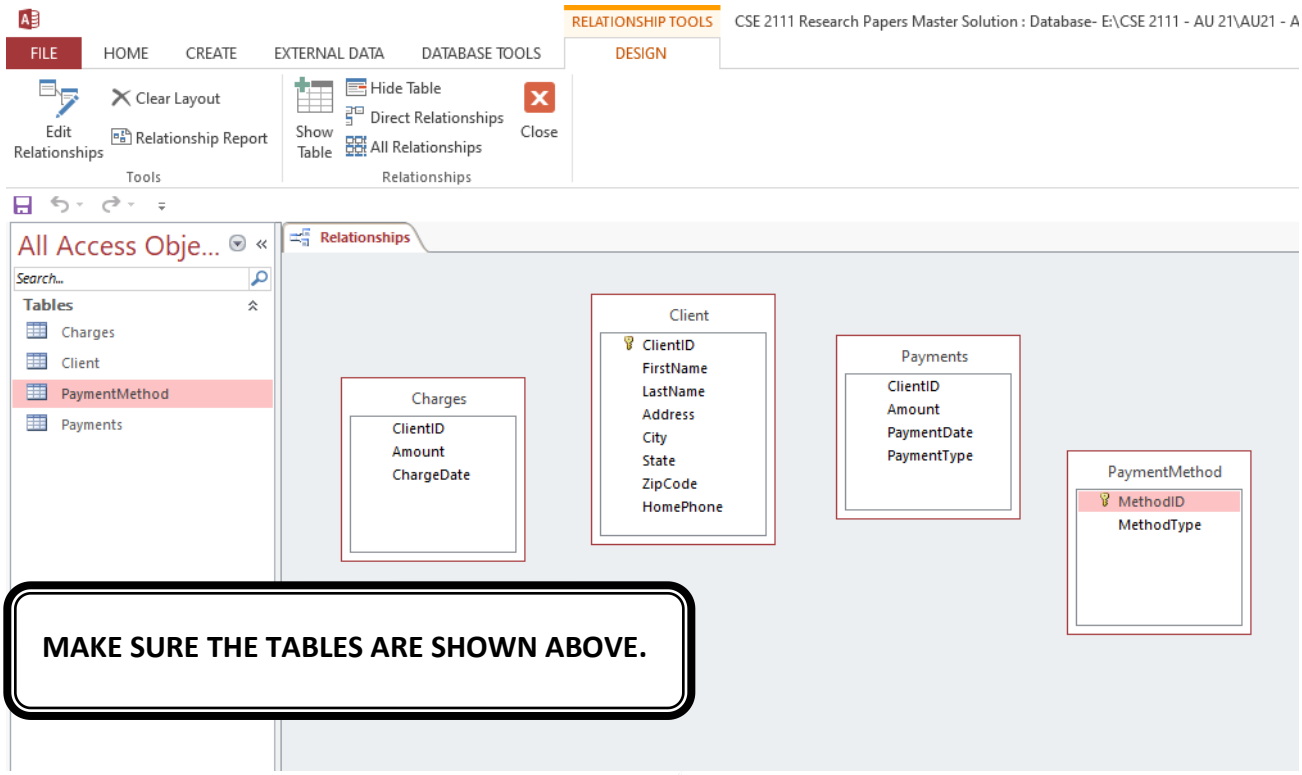
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2. Double Click on the Charges table to add the table to the relationships window. Double click on the remaining tables to add them to the relationships window and click on the Close button.



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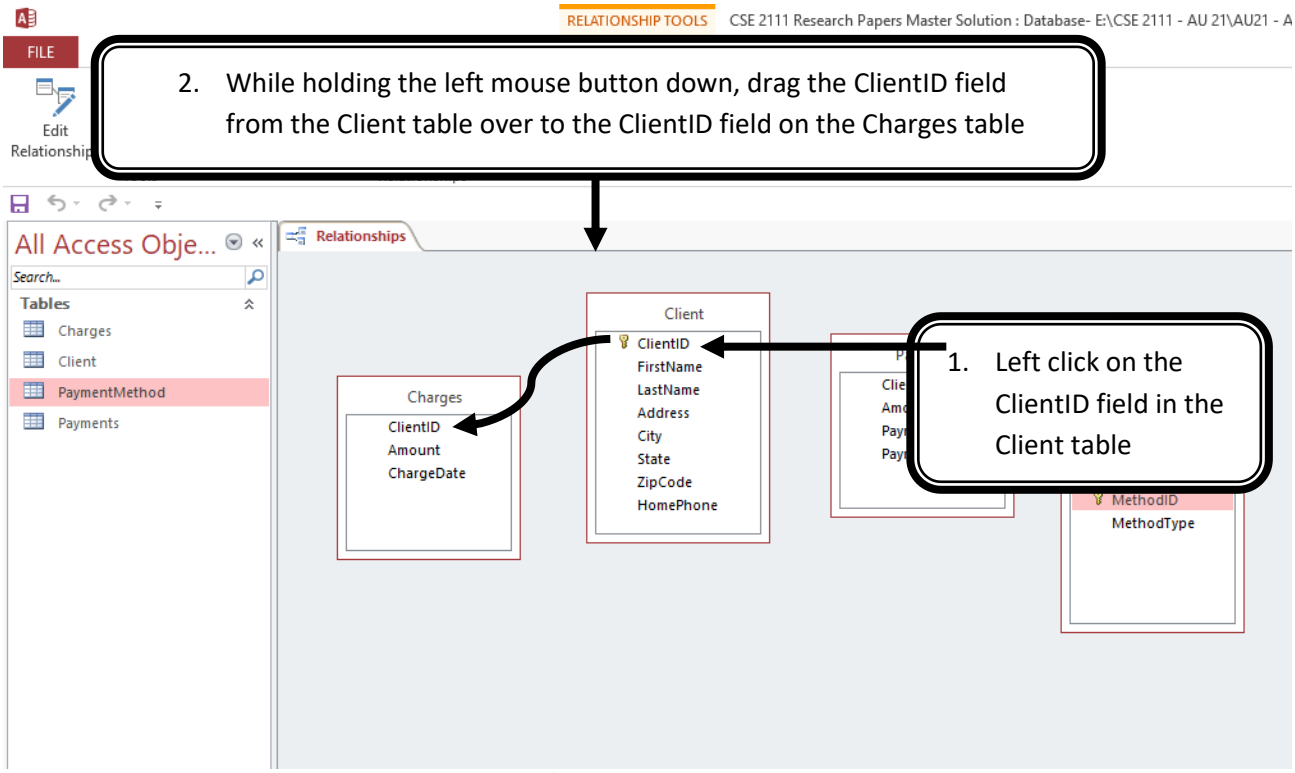
3. Arrange the tables on the screen as shown. (You can drag them by clicking the left mouse button.)



MAKE SURE THE TABLES ARE SHOWN ABOVE.

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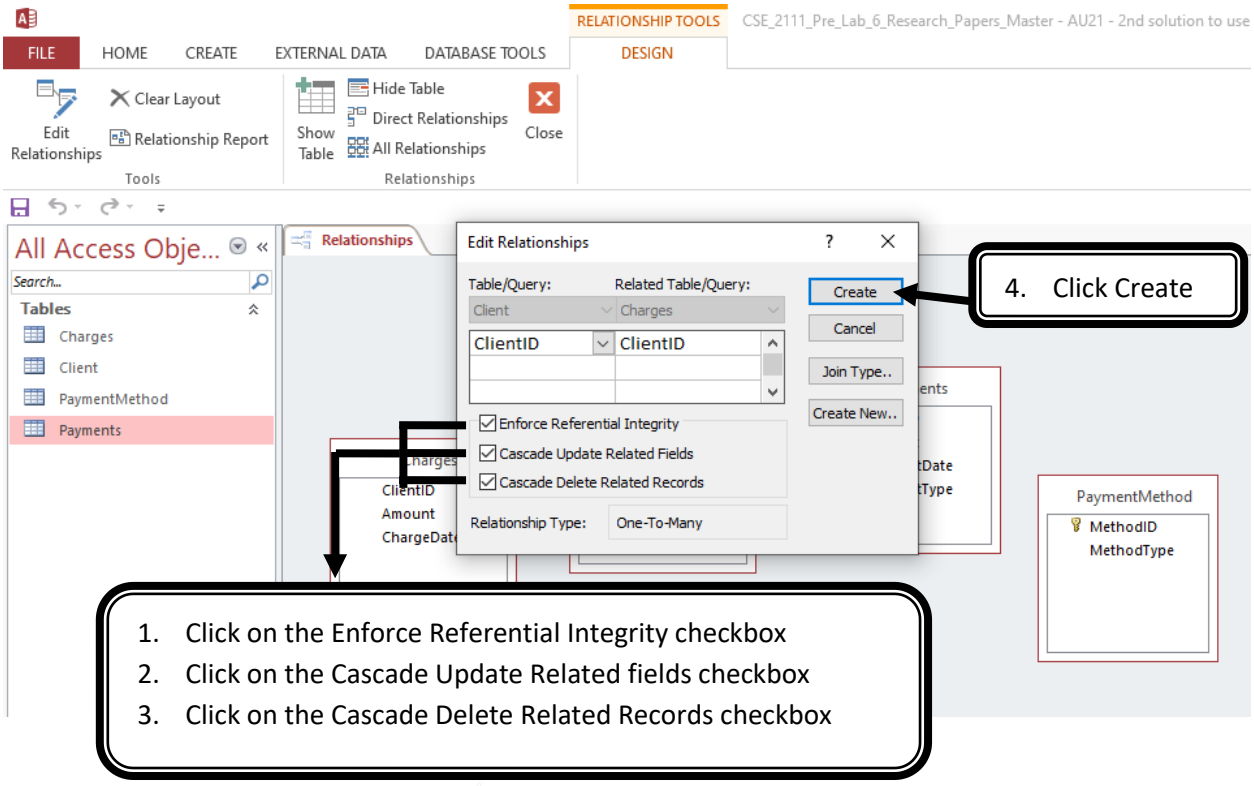
4. Click on the left mouse button on the ClientID field in the Client table. While holding down the left mouse button drag the field over to the ClientID in the Charges table. You will have a relationship dialog box pop up as shown below.



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5. Click the Enforce Referential Integrity, Cascade Update Related fields, and Cascade Delete Related Records check boxes and then click the Create button.

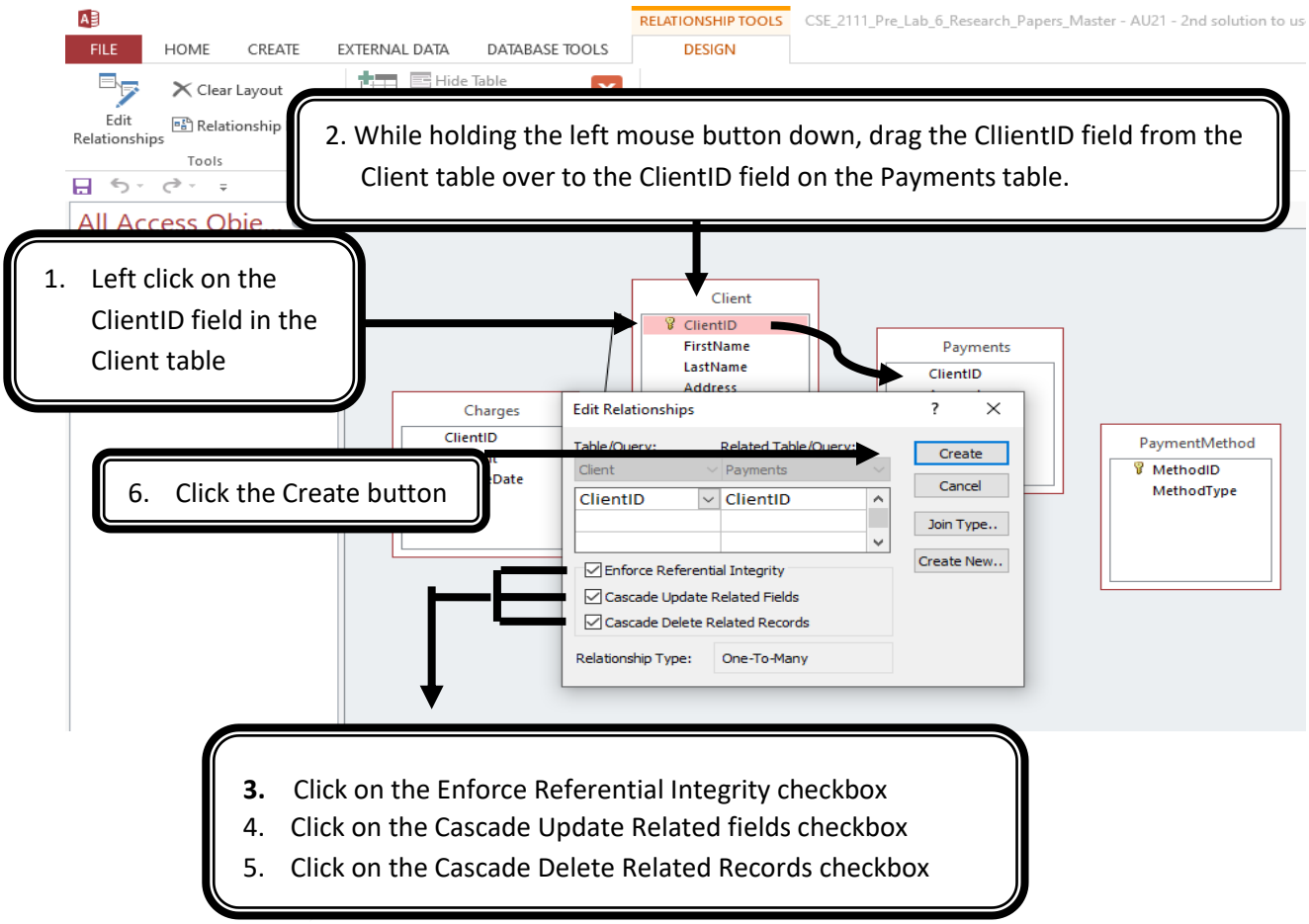
*****NOTE** We will discuss later the importance of the Cascade Update Related Fields and Cascade Delete Related Fields Check box.***



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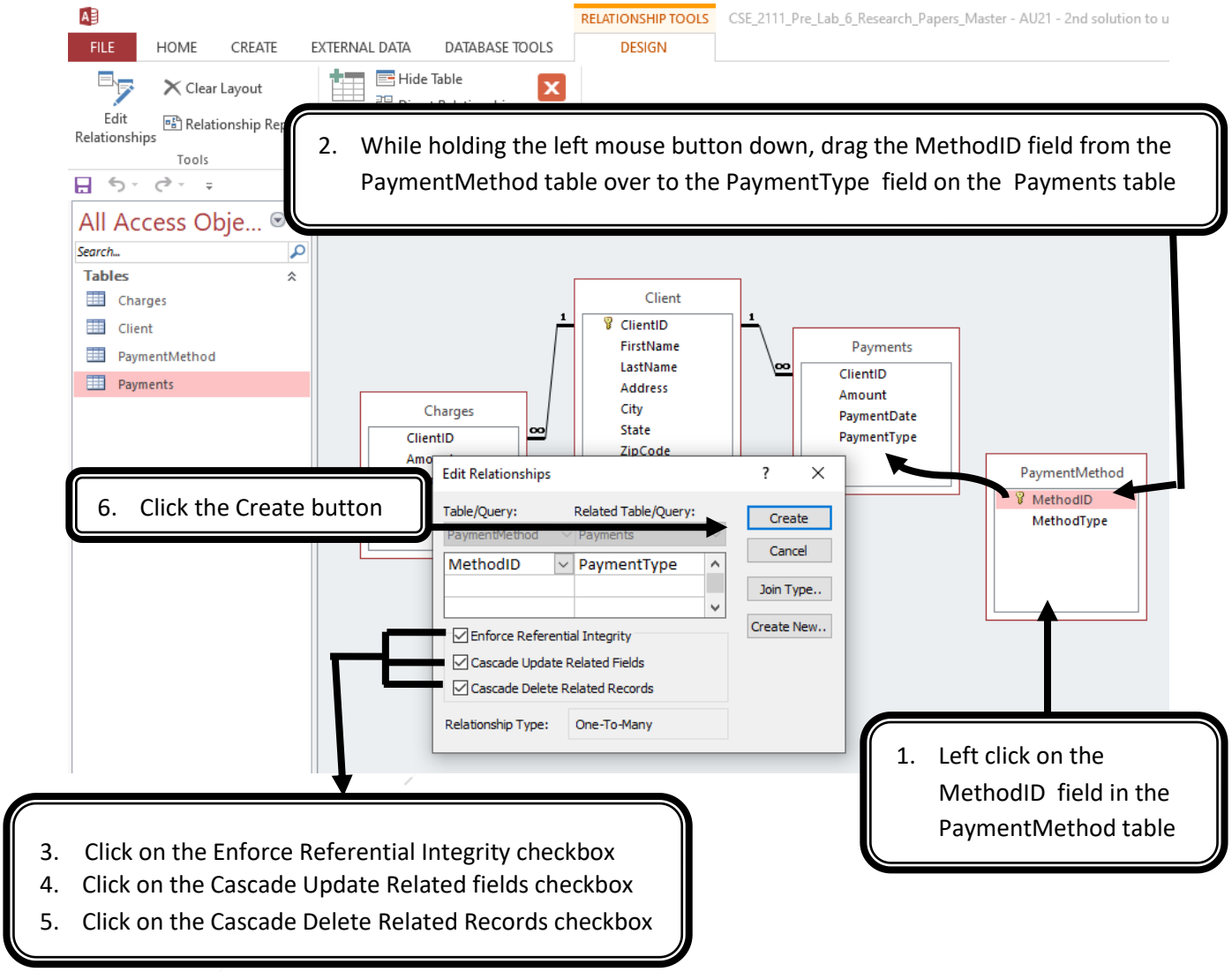
6. Click the left mouse button on the ClientID field in the Client table and drag it over the ClientID in the Payments table. You will have a relationship dialog box pop up as shown below. Click on the Enforce Referential Integrity, Cascade Update Related fields, and Cascade Delete Related Records check boxes and then click the Create button.

***NOTE** We will discuss later the importance of the Cascade Update Related Fields and Cascade Delete Related Fields Check box.*



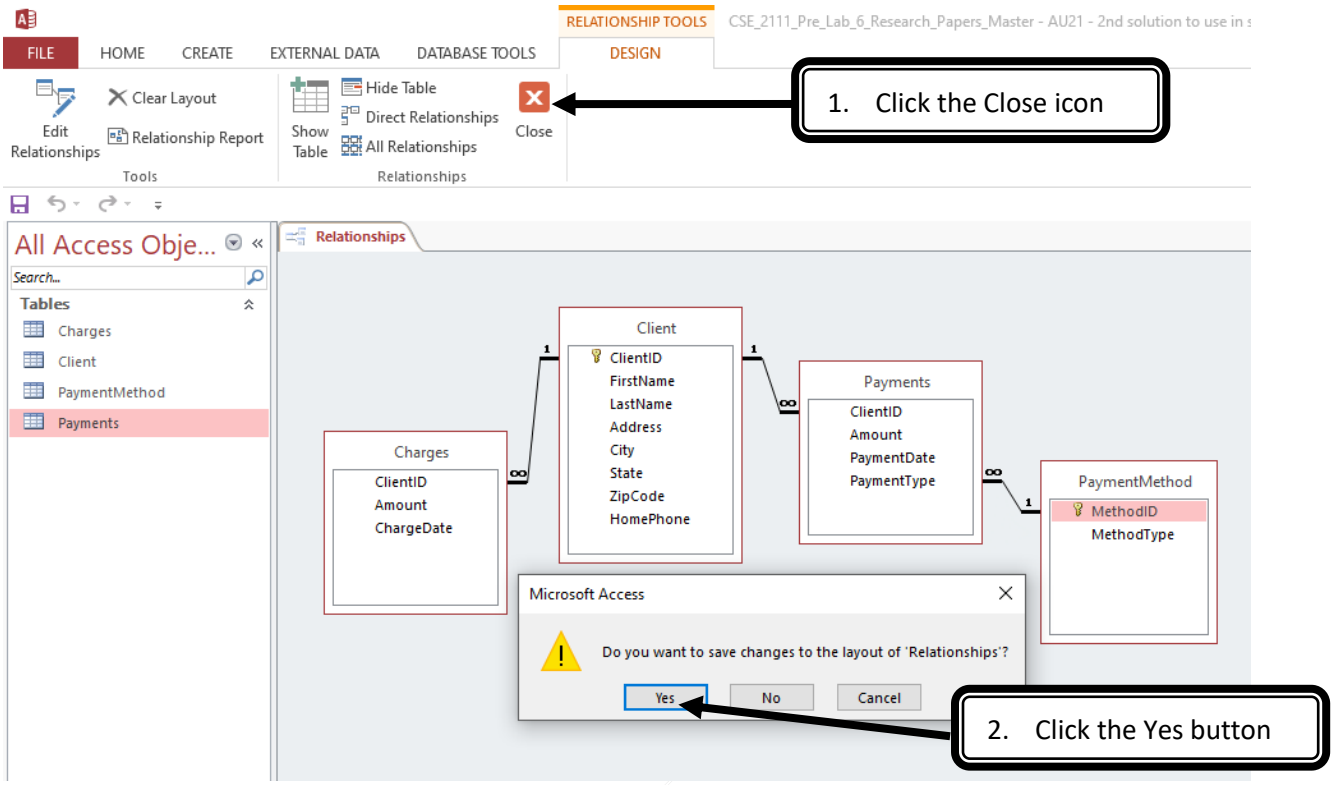
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7. Click the left mouse button on the MethodID field in the PaymentMethod table and drag it over to the PaymentType field in the Payments table. You will have a relationship dialog box pop up as shown below. Click on the Enforce Referential Integrity, Cascade Update Related fields, and Cascade Delete Related Records check boxes and then click the Create button.



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8. Close and save the relationships.

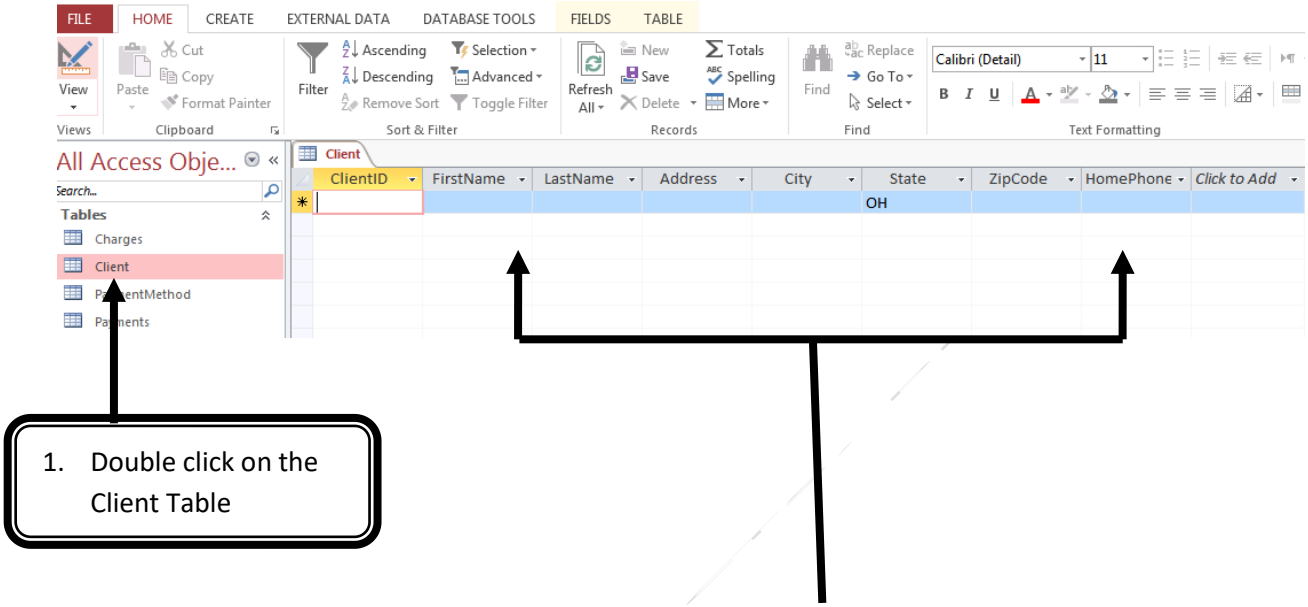


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STEP 8 – POPULATE THE DATABASE.

Now that you have created the structure of the database, it is time to input your client information.

1. Double click on the Client Table to open it up in **Datasheet View** and type in the information for the following records.



1. Double click on the Client Table

NOTE

- * indicates a blank record where you can input information.
  - This row will always show up. It is not considered a record. It is just a holding place to add new records.
- OH is the default for the state field.
  - It will always show up at first, but you can change the OH to any other state you wish.
- When you type in the information, don't worry if it doesn't display completely in the cell. It is there, you just can't see it.

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2. ***Important***: Type the first 5 records from the Client table shown in *Figure 1.6* (for ClientID J7500 through J7520). Then, you will use the Import feature to complete the Client table information.

Client Table

ClientID	FirstName	LastName	Address	City	State	ZipCode	HomePhone
J7500	Nancy	Davolio	507 20th Ave. E. Apt. 2A	Seattle	WA	98122	(614)-292-3900
J7505	John	Smith	123 Main Street	Columbus	OH	43081	(740)-365-9192
J7510	Mark	Johnson	420 Broad Street	Columbus	OH	42001	(740)-365-7700
J7515	Jen	Griffin	500 High Street	Columbus	OH	43081	(614)-292-4566
J7520	Jennifer	King	400 Walnut Street	Cincinnati	OH	41098	(614)-888-9930
Import the following data into your database Many times, the data you would like to include in your database will come from an outside source such as a text file, xml file, Excel file, or even another database file. If you set up this file correctly, you can import the data from the file into your database tables. You will import the records shown below. Follow the steps after Figure 1.6 to accomplish this task.							
J7525	Karen	Day	250 Fifth Street	Cincinnati	OH	41077	(614)-293-5001
J7530	Jeff	Blair	125 Plum Street	Cincinnati	OH	41077	(740)-365-4478
J7535	Nancy	Beard	10 South Wacker	Chicago	IL	60639	(740)-365-4879
J7540	Brad	Wallace	120 South LaSalle	Chicago	IL	60639	(740)-365-4421
J7545	Joe	Davis	15 West Washington	Chicago	IL	60221	(614)-443-0988
J7550	Tim	Smith	908 W. Capital Way	Tacoma	WA	98401	(614)-293-9655
J7555	Brandon	Coake	722 Moss Bay Blvd.	Kirkland	WA	98033	(614)-555-9333
J7560	Margaret	Peacock	4110 Old Redmond Rd.	Redmond	WA	98052	(614)-293-8857
J7565	Matthew	Dunn	14 Garrett Hill	Seattle	WA	98105	(740)-365-4433
J7570	George	Nagy	722 DaVinci Blvd.	Kirkland	WA	98034	(740)-365-2241
J7575	Deborah	Peterson	305 14th Ave. S. Suite 3B	Seattle	WA	98128	(740)-365-9031

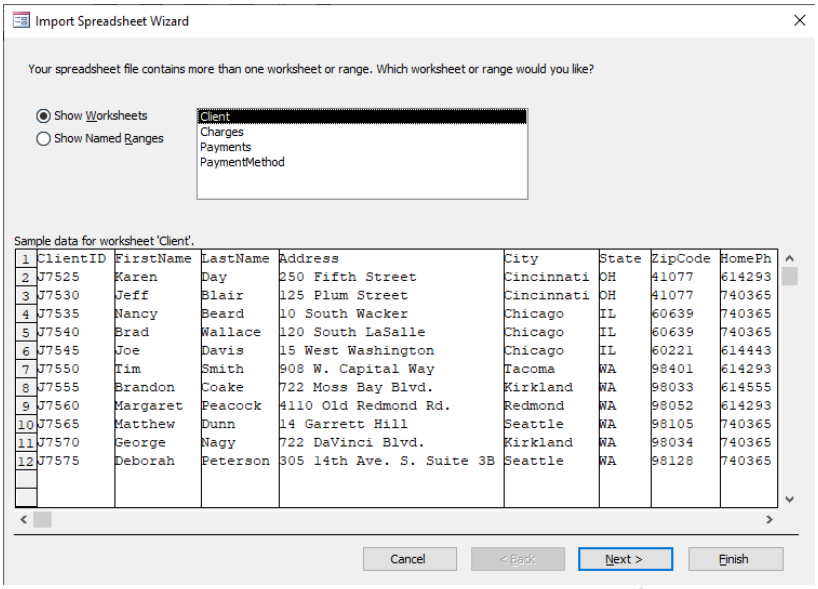
Figure 1.6

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Import the information into the database from an Excel file.

We could input all the information, but I have input the data in an Excel file, and I am going to have you import (insert the data) in the respective tables in the database.

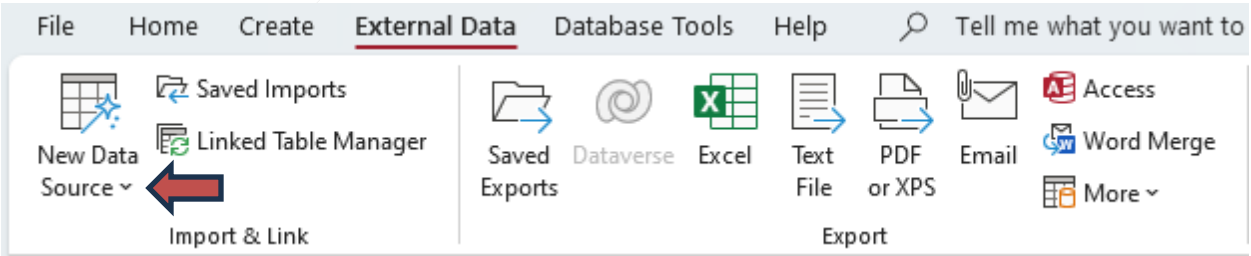
NOTE: The table names and field names must be spelled exactly as the names in the database tables, or the Import will not work correctly. (There should be no spaces in the field names, and you must use the proper upper/lower case when creating the field names.)



	ClientID	FirstName	LastName	Address	City	State	ZipCode	HomePh
2	J7525	Karen	Day	250 Fifth Street	Cincinnati	OH	41077	614293
3	J7530	Jeff	Blair	125 Plum Street	Cincinnati	OH	41077	740365
4	J7535	Nancy	Beard	10 South Wacker	Chicago	IL	60639	740365
5	J7540	Brad	Wallace	120 South LaSalle	Chicago	IL	60639	740365
6	J7545	Joe	Davis	15 West Washington	Chicago	IL	60221	614443
7	J7550	Tim	Smith	908 W. Capital Way	Tacoma	WA	98401	614293
8	J7555	Brandon	Coake	722 Moss Bay Blvd.	Kirkland	WA	98033	614555
9	J7560	Margaret	Peacock	4110 Old Redmond Rd.	Redmond	WA	98052	614293
10	J7565	Matthew	Dunn	14 Garrett Hill	Seattle	WA	98105	740365
11	J7570	George	Nagy	722 DaVinci Blvd.	Kirkland	WA	98034	740365
12	J7575	Deborah	Peterson	305 14th Ave. S. Suite 3B	Seattle	WA	98128	740365

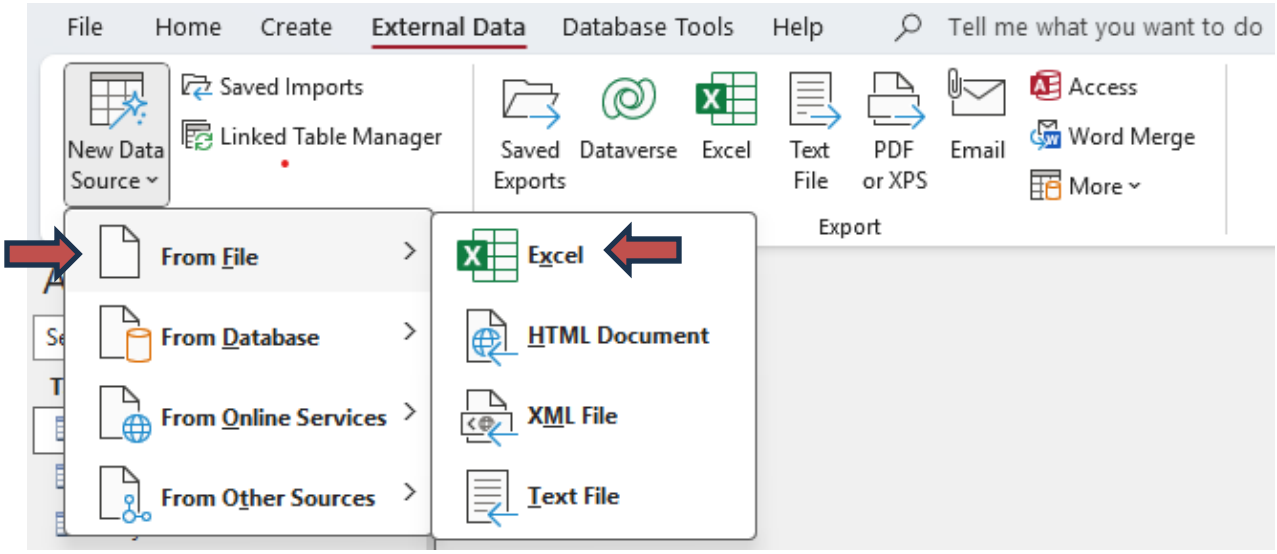
1. Close all the tables before you continue.

2. Click on the External Data tab, then click on the **New Data Source in the Import & Link group**. **DO NOT CHOOSE the Excel Icon under the Export Tab. You do NOT need to save the Import Steps.**



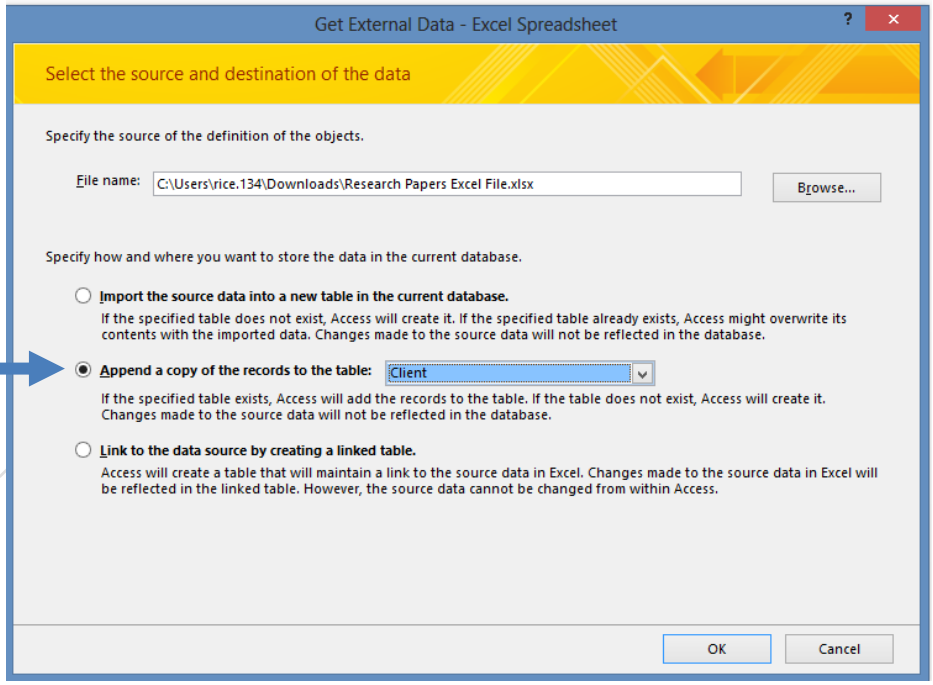
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3. Click on From File. Then choose the type of file you are importing. For Research Papers, it is an Excel file.



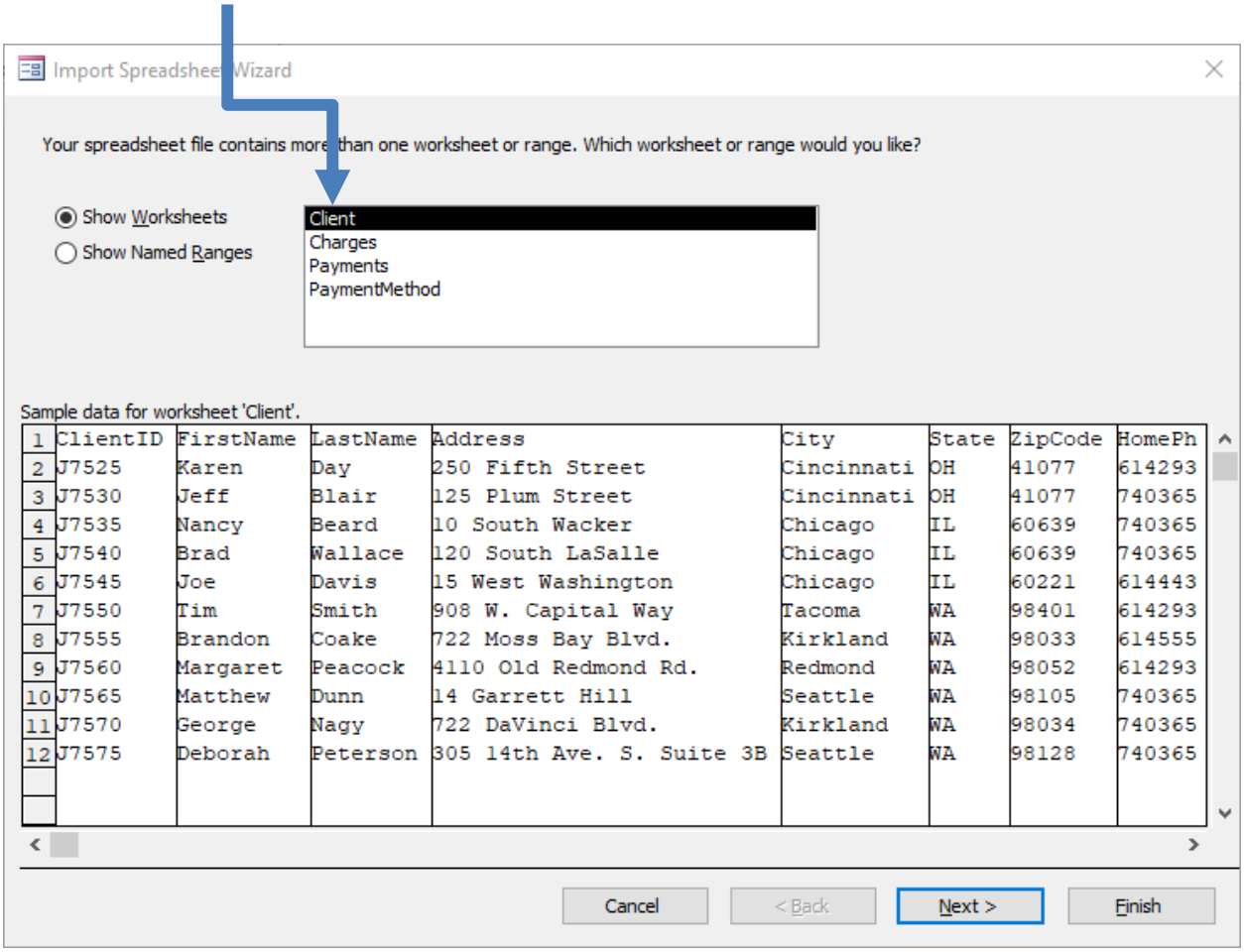
4. Browse to the location of Research Papers Excel File.xlsx, click the **Append a copy of the records to the table** option button and select the ***Client*** table. Click OK.

5. Click OK.



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6. Select the Client table and click Finish.



7. Open the Client table. Notice, you have populated the Client table with the data in your Excel file. **You should now see 16 records in the Client Table. Reminder: If you did NOT type in the first 5 records that are listed at the top of Figure 1.6, you will not have 16 records in the Client Table. You should have 16 total records in the Client Table after the Import process.**
8. Close the Client table.
9. Using the Research Papers Excel File.xlsx file, **(IMPORTANT):** import the remainder of the tables in the **following order:**

   a. **Payment Method**
   b. **Payments**
   c. **Charges**

10. **Your 4 tables should now be populated: Client Table with 16 Records, PaymentMethod with 5 Records, Payments with 11 Records, and Charges with 23 Records.**
11. **Save the Access Database, close it, and you will upload this file to Simnet (Pre-Lab 6 Research Papers Master) or wait until the pre-lab is completed and upload all files to Simnet at one time.**

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When you import dates, they may display as numbers, but they will convert to dates when you import the fields.

CASCADE UPDATE RELATED RECORDS/CASCADE DELETE RELATED RECORDS

If you recall, when we enforced referential data integrity in MS Access, we checked two boxes to Cascade Update Related Records and Cascade Delete Related Records. These options ensure that our database has no orphans. If you remember correctly, when we enforce referential data integrity the database will not allow any orphans to be input in the tables. But orphans can still exist in your database even if you click the enforce referential data integrity checkbox. For example, if you look in the database you will notice Nancy Davolio, ClientID J7500, has 2 records in the Charges table, and 3 records in the Payments table. What would happen if I deleted the record for Nancy Davolio in the Client table? Now there are 2 orphans in the Charges table, and 2 orphans in the Payments table. Likewise, if I change the ClientID for Nancy Davolio to J7900, I also produce orphans in both the Charges table and the Payments table. However, if I click the checkboxes to Cascade Update Related Records, and Cascade Delete Related Records, if I delete the parent, Nancy Davolio, ClientID J7500, I will delete all the J7500 children in all of the related tables. But, if I delete the children, J7500, in the Charges or Payments table, I will not delete the parent J7500 in the Client table. The Update Related Records checkbox works the same way as the Cascade Delete Related Records, only it updates the children's ClientID when the parent ClientID is updated. Remember, it only goes one way. If I delete the parent, I delete the children, but if I delete the children, I do not delete the parent. If I update the ClientID for the parent, the ClientID for all the related children will be updated, but if I update the ClientID for the children table, the ClientID for the parent table will not be updated.

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Possible Scenarios when the following boxes are checked when creating relationships:

Enforce Referential Integrity, Cascade Update Related Fields, Cascade Delete Related Records

NOTE: DO NOT MAKE THESE CHANGES IN THE DATABASE

THESE ARE EXAMPLES ONLY

Action	**Associated Action**
Delete record J7500 in the Client table	All records in the Charges table with ClientID J7500 will be deleted. All records in the Payments table with ClientID J7500 will be deleted.
Delete any J7500 record in the Charges Table	Only that one J7500 record will be deleted in the Charges Table
Delete any J7500 record in the Payments Table	Only that one J7500 record will be deleted in the Payments Table
Change ClientID J7500 to J9900 in the Client table	All records in the Charges table with ClientID J7500 will be changed to ClientID J9900 All records in the Payments table with ClientID J7500 will be changed to ClientID J9900
Delete record with MethodID of 1 in the PaymentMethod table.	All records in the Payments table with PaymentType of 1 will be deleted.
Change record with MethodID 1 to MethodID 6 in the PaymentMethod table.	All records in the Payments table with PaymentType of 1 will be changed to PaymentType of 6.
Change record J7500 from PaymentType of 1 to PaymentType of 6 in the Payments table.	If a record with MethodID 6 exists in the PaymentMethod table the record in the Payments table with will change accordingly. If a record with MethodID 6 does not exist in the PaymentMethod table, you will receive the message, “Referential Data Integrity has been Violated”, and you will not be able to make the change to the record.

PART 2: DATABASE CONCEPTS

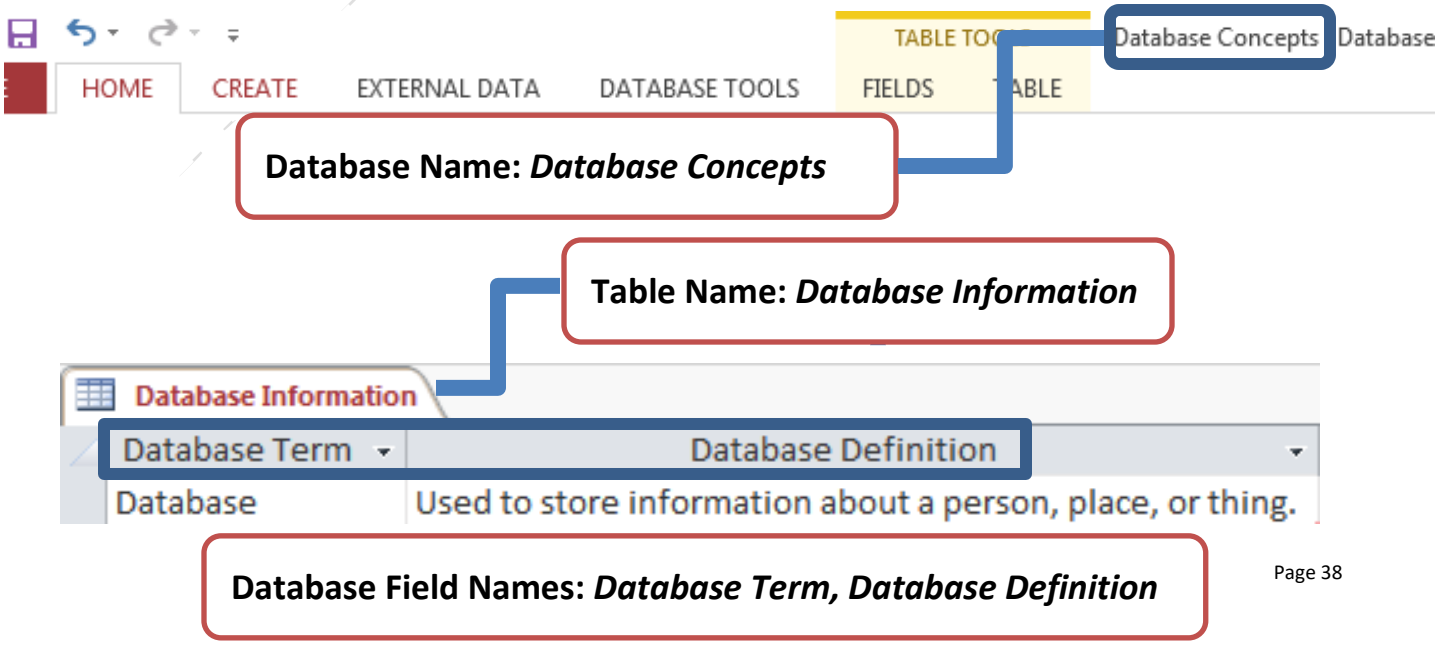
1. **Open the Access database named, CSE 2111 Pre Lab 6 Database Concepts Master.accdb.**
2. **Notice that a Table has already been created for you that is named *Database Information*.**
3. **Click on the Design View button and you will see that the Database Table Structure has been created: two fields, field types, field sizes, and field properties have been set up. The Long Text Field Size supports a large amount of data entry to this field.**

Table Name:	Database Information	Field Type	Field Size (Properties)	
Fields:	Database Term	Short Text	255	This table does not have a primary key.
	Database Definition	Long Text		

4. **The following database terms (shown below) have been provided for you.**
5. **Locate the term definitions in your Chapters 10 and 11 Course Notes and ppts and Input them into the *Database Definition* field.** *(See example below.)*
6. **Once you have filled in the definitions, save and close the Access file, and upload file to Simnet.**

Steps when creating a database/short synopsis	Foreign key
Data	Database
Table	Record
Data type/Field type	Primary Key
DBMS	Referential Data Integrity
Report	Requirements of a valid foreign key
Structured Query Language	Hierarchy of data in a database
Data sheet View	Dynaset
Information	Cascade Updated records
QBE grid	Cascade delete records
Fields	

Example



PART 3: CTC CONSULTING

CTC Consulting is a company that provides extensive training in the MS Office Suite. The company offers beginning, intermediate, and advanced classes. The company would like you to create a database to track their business. Note: The instructors at CTC are treated as independent contractors, meaning they are not employed by the company, and are paid for each class they teach. Based on the company's needs, you have decided on the following database structure.

- **Customer:** This table will contain information about each customer such as Name, Address, Phone Number, and if the customer currently receives the company's emails detailing the training offered during the upcoming months. Each customer is uniquely identified by a Customer ID.
- **Class:** This table contains the relevant information for each class that is offered by CTC, including class name, meeting times, cost of the class per student, and other information. Each class is represented by a unique ClassID and also contains the InstructorID of the instructor that currently teaches the class.
- **ClassEnrollment:** This table lists the information for every course registration. It includes the Customer's ID, Class ID (the course they registered for), and whether they have paid the fees for this course.
- **Instructor:** This table contains the relevant information for the instructors. It contains their name, Address, Contact Information, and instructor charges Antiques Mall per class. Each instructor is given a unique Instructor ID.

1. **Open the database, CSE 2111 Pre Lab 6 CTC Consulting Master.accdb**
2. **Create the database structure shown on the next page.**
3. **Set the primary keys, create the table relationships, and enforce referential data integrity. Note: you will need to complete this step before you import data.**

Screenshot:

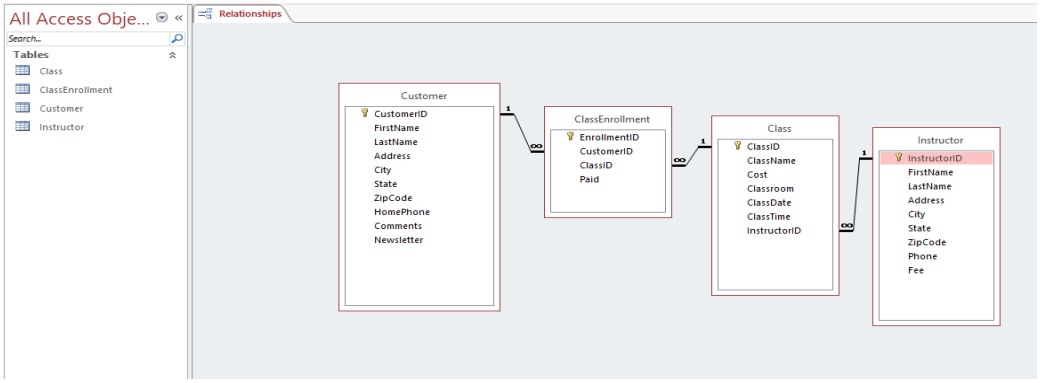
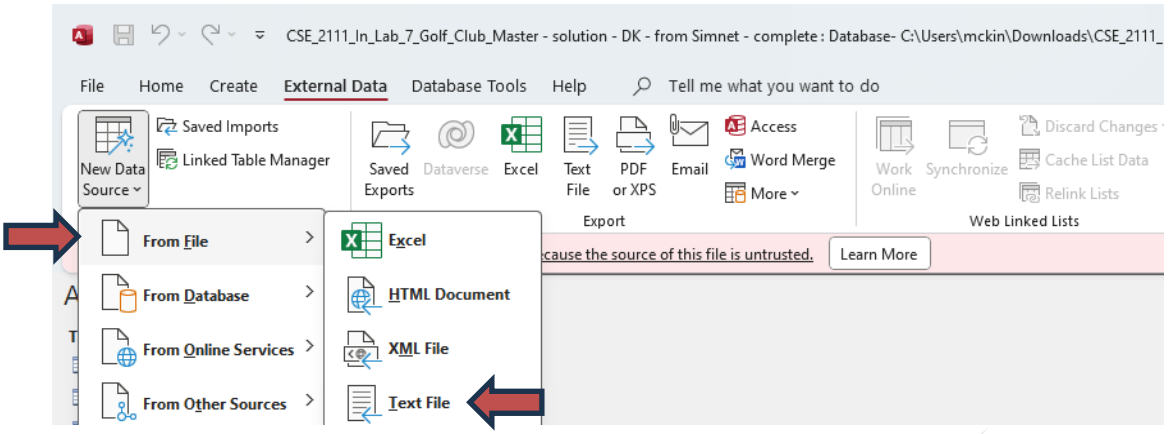


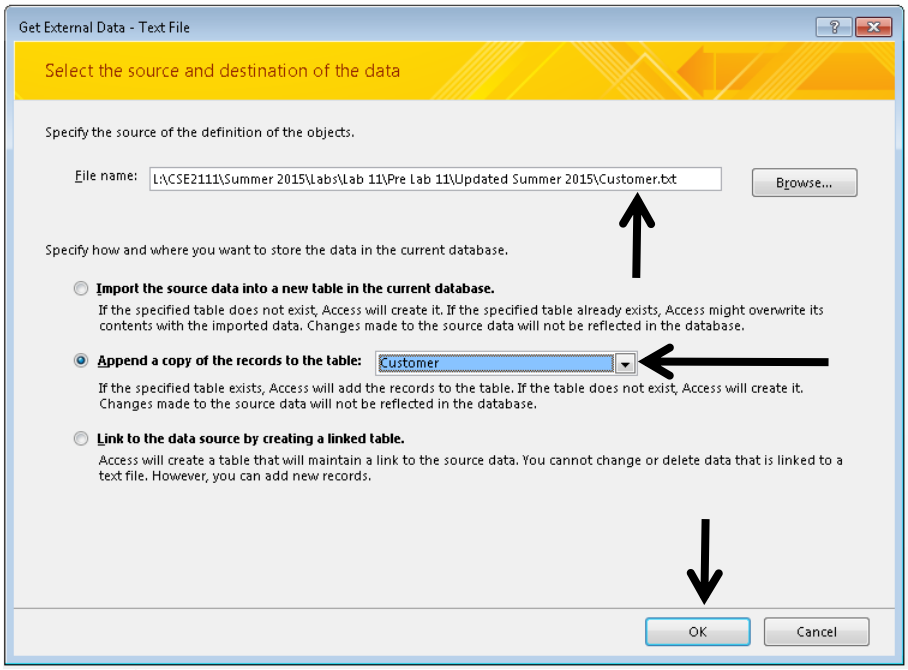
Table Name:	Customer	Field Type	Properties	
Fields:	CustomerID	Short Text	10 Characters long	Primary Key
	FirstName	Short Text	25 Characters long	
	LastName	Short Text	50 Characters long	
	Address	Short Text	60 Characters long	
	City	Short Text	25 Characters long	
	State	Short Text	2 Characters long	"OH" as default value
	ZipCode	Short Text	5 Characters long	
	HomePhone	Short Text	10 Characters long	Input Mask
	Comments	Short Text	150 Characters long	
	Newsletter	Yes/No	None	
Table Name:	**ClassEnrollment**	**Field Type**	**Properties**	
	EnrollmentID	Short Text	10 Characters long	Primary Key
Fields:	CustomerID	Short Text	10 Characters long	
	ClassID	Short Text	10 characters long	
	Paid	Yes/No	None	
Table Name:	**Class**	**Field Type**	**Properties**	
Fields:	ClassID	Short Text	10 Characters long	Primary Key
	ClassName	Short Text	50 Characters long	
	Cost	Currency	None	
	Classroom	Short Text	10 Characters long	
	ClassDate	Date/Time	Format: Long Date	
	ClassTime	Date/Time	Format: Medium Time	Input Mask
	InstructorID	Short Text	10 Characters long	
Table Name:	**Instructor**	**Field Type**	**Properties**	
Fields:	InstructorID	Short Text	10 Characters long	Primary Key
	FirstName	Short Text	25 Characters long	
	LastName	Short Text	50 Characters long	
	Address	Short Text	60 Characters long	
	City	Short Text	25 Characters long	
	State	Short Text	2 Characters long	"OH" as default value
	ZipCode	Short Text	5 Characters long	
	Phone	Short Text	10 Characters long	Input Mask
	Fee	Currency	None	

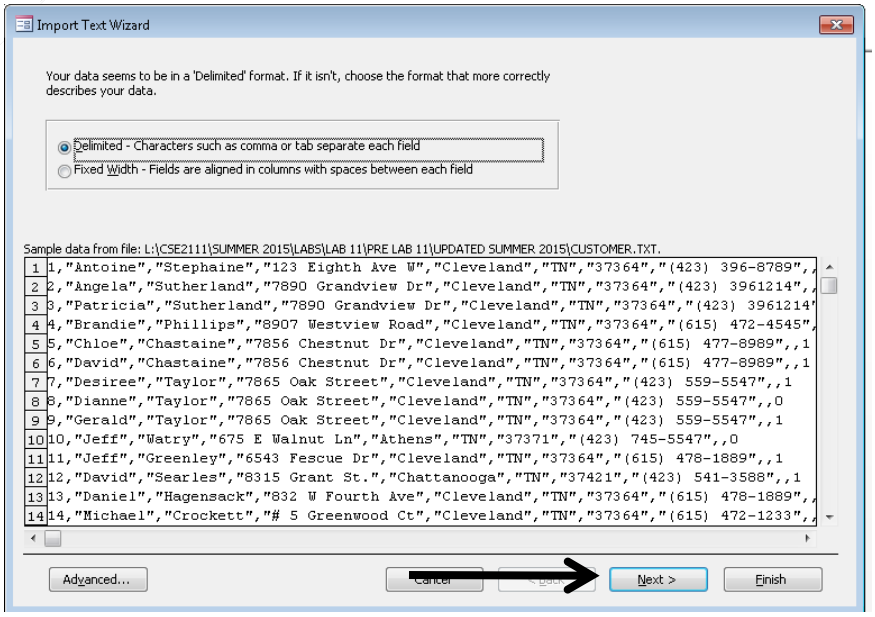
In part 1 of this lab, you populated the database by using the import wizard to import data from an **Excel file** into the respective tables. In this part of the lab, you will populate the database using the import wizard to import data from a **text file**.

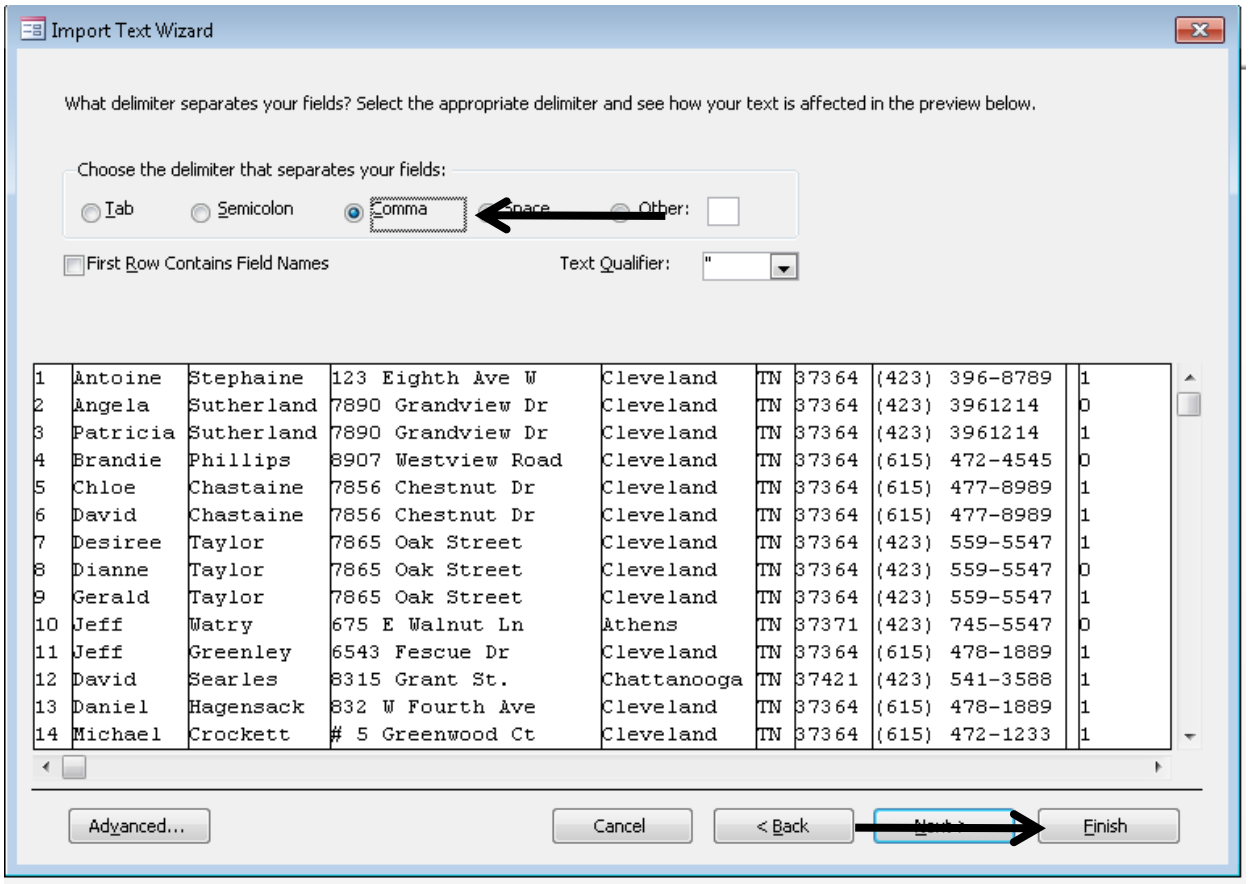
Follow the instructions below to import your four text files into your tables. You do NOT need to save your imports each time. Import the files in exactly the order described and into the correct tables. Once you have done the four imports, all your tables should be populated with the data from the text files.

4. Import the data in the Customer.txt file.









5. **Import the remainder of the data in the following order.**
    a. **Instructor**
    b. **Class**
    c. **Class Enrollment**

6. **Double click on each Table name to see the imported data.**
7. **Close all tables, save if necessary, and upload your three databases to Simnet.**

Lab submissions

1. **CSE 2111 Pre Lab 6 Research Papers Master.accdb**
2. **CSE 2111 Pre Lab 6 Database Concepts Master.accdb**
3. **CSE 2111 Pre Lab 6 CTC Consulting Master.accdb**